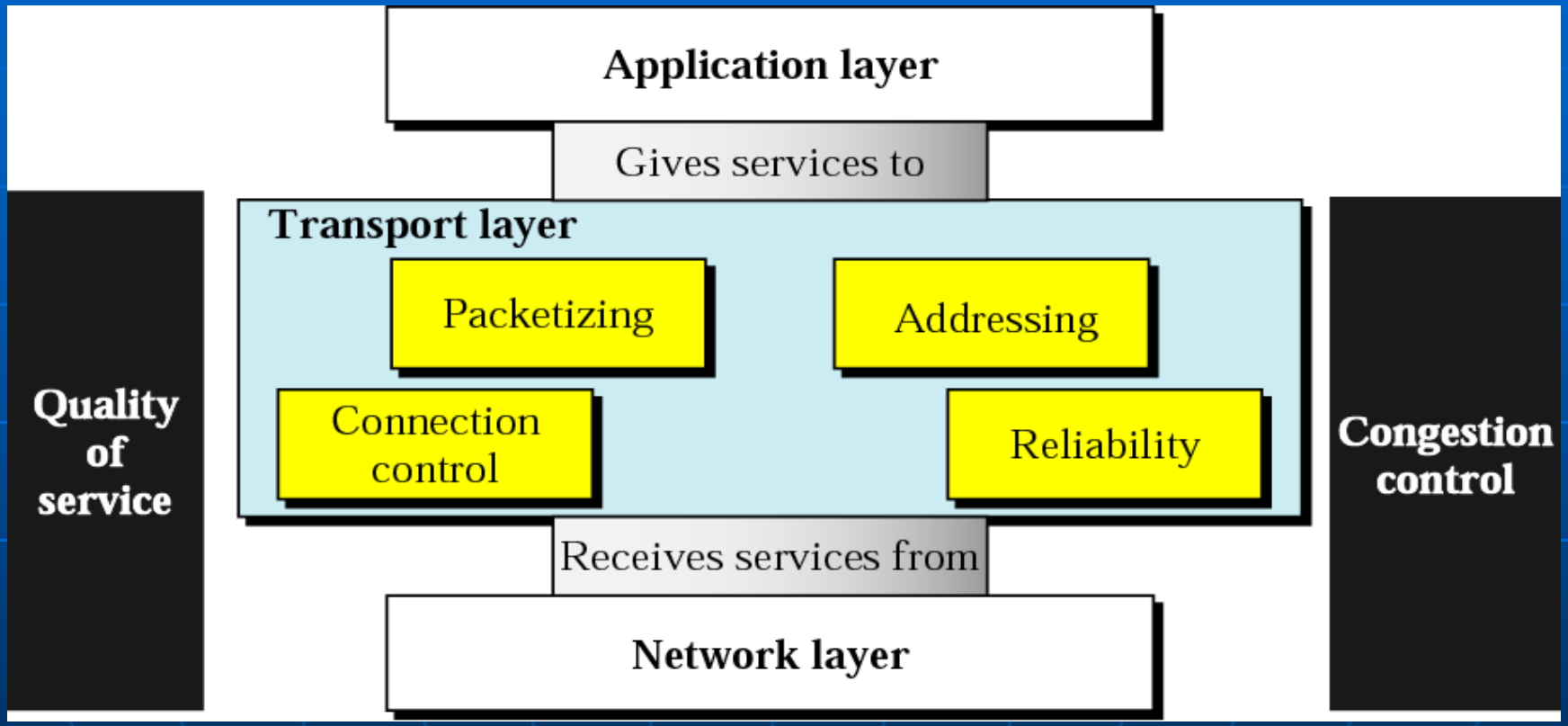
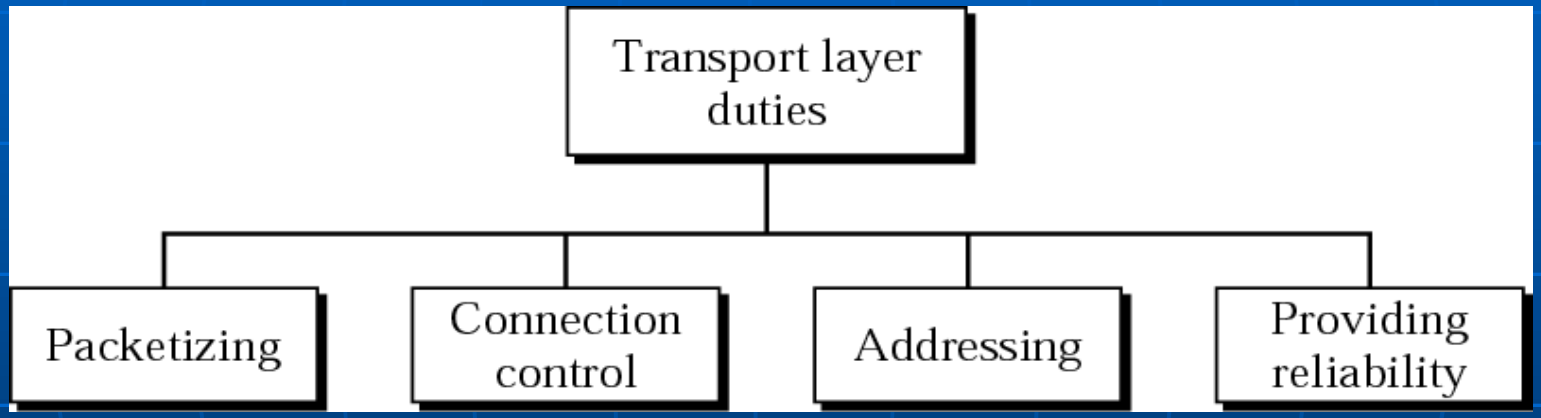
Transport Layer

Position of transport layer



Transport layer duties



Chapters

Chapter 22 ***Process-to-Process Delivery***

Chapter 23 ***Congestion Control and QoS***

Chapter 22

Process-to-Process Delivery:

UDP and TCP

22.1 Process-to-Process Delivery

Client-Server Paradigm

Addressing

Multiplexing and Demultiplexing

Connectionless/Connection-Oriented

Reliable/Unreliable



Note:

The transport layer is responsible for process-to-process delivery.

Figure 22.1 Types of data deliveries

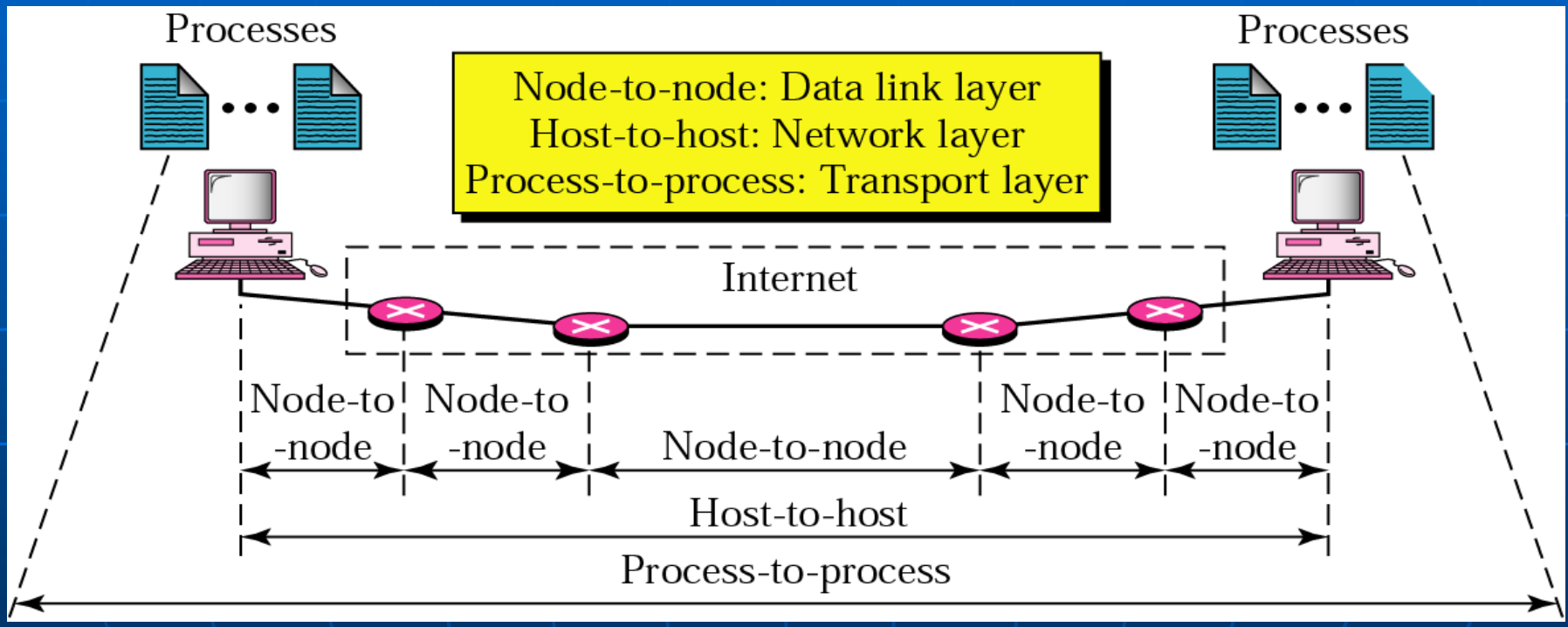


Figure 22.2 Port numbers

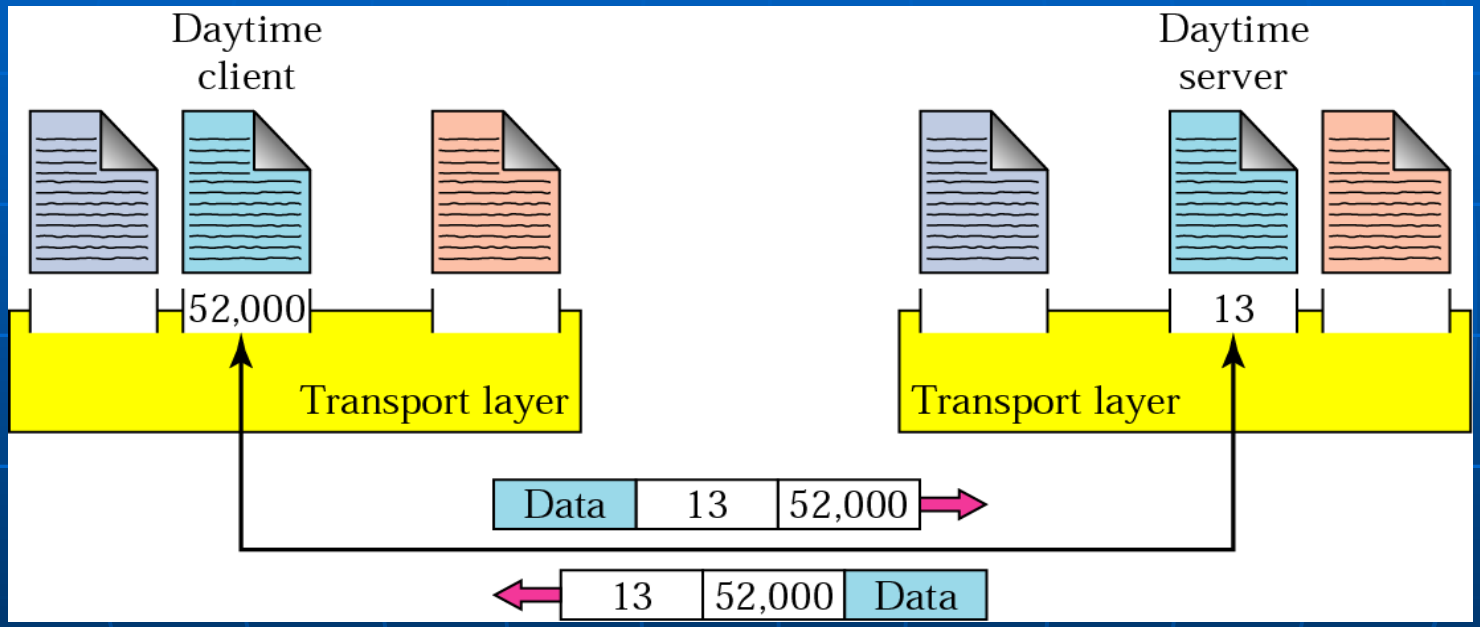


Figure 22.3 IP addresses versus port numbers

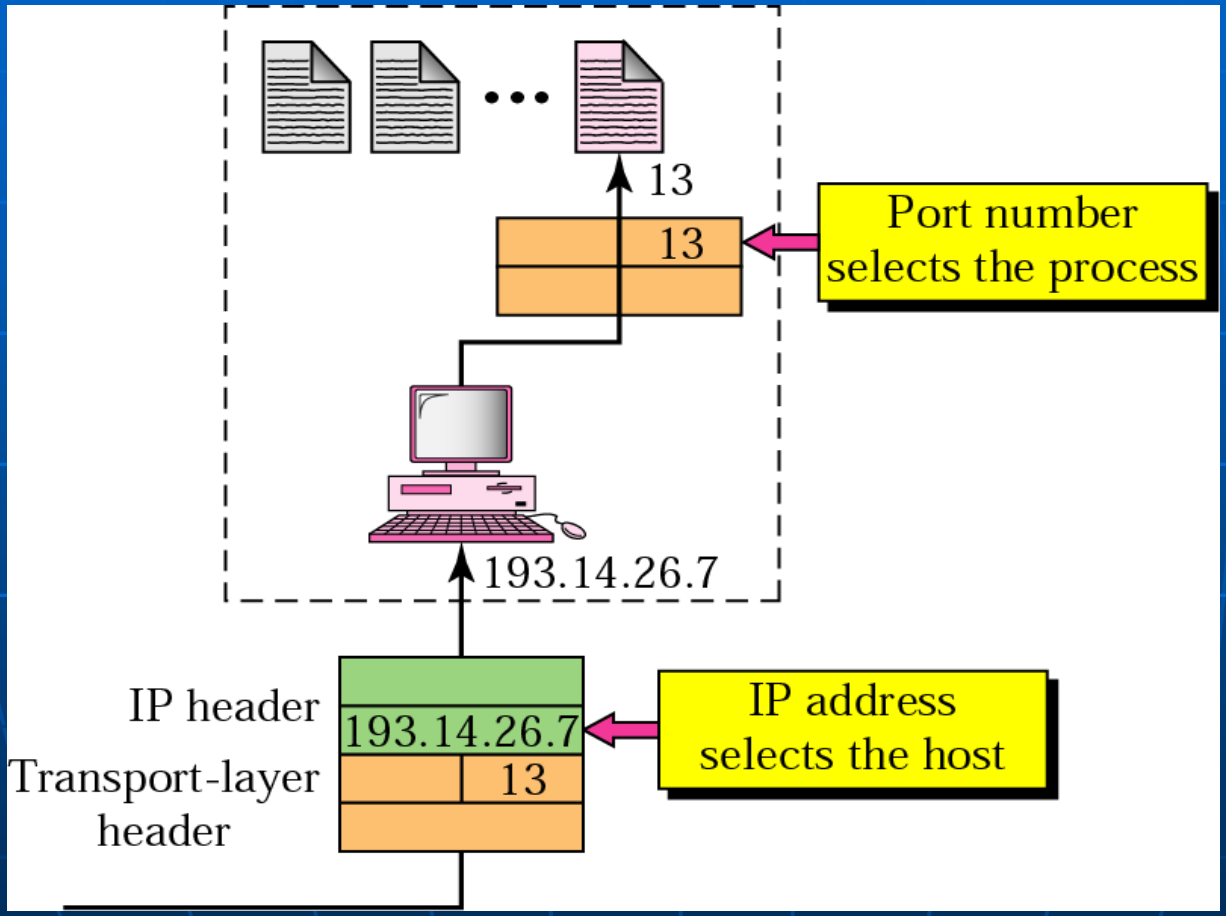


Figure 22.4 IANA ranges

Internet Assigned Numbers Authority

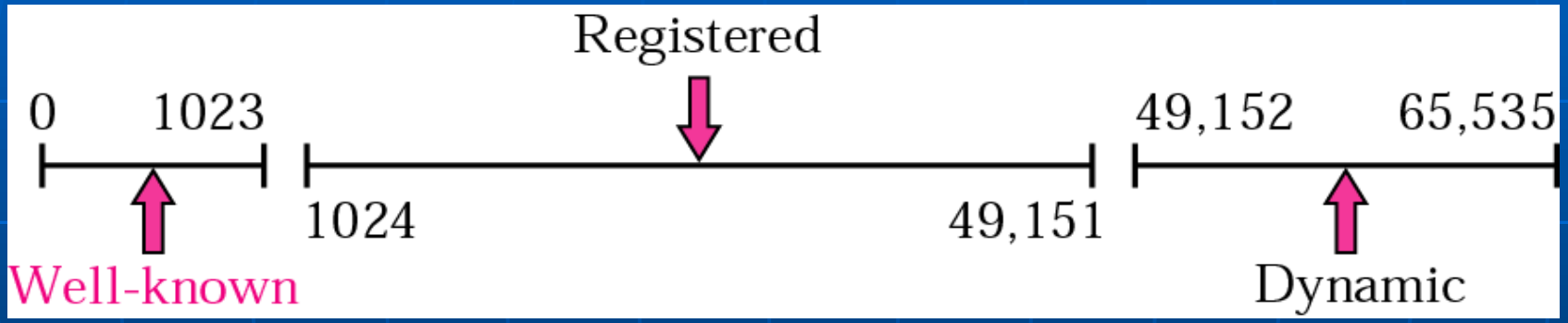


Figure 22.5 Socket address

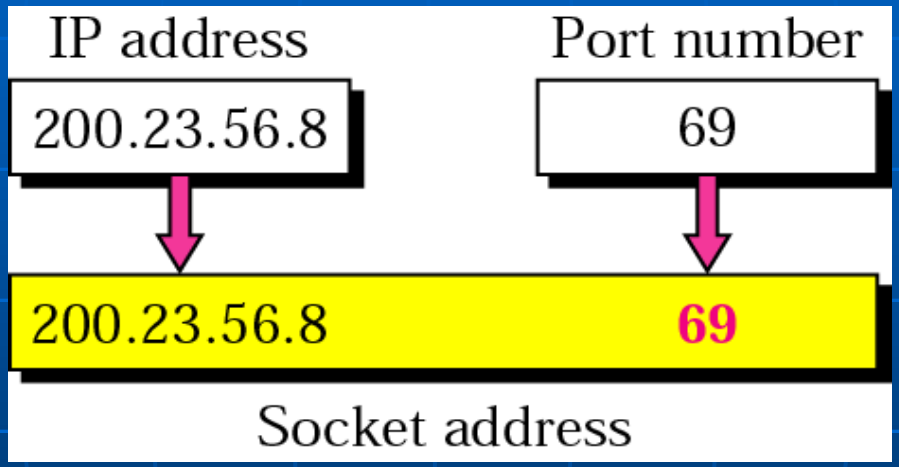


Figure 22.6 Multiplexing and demultiplexing

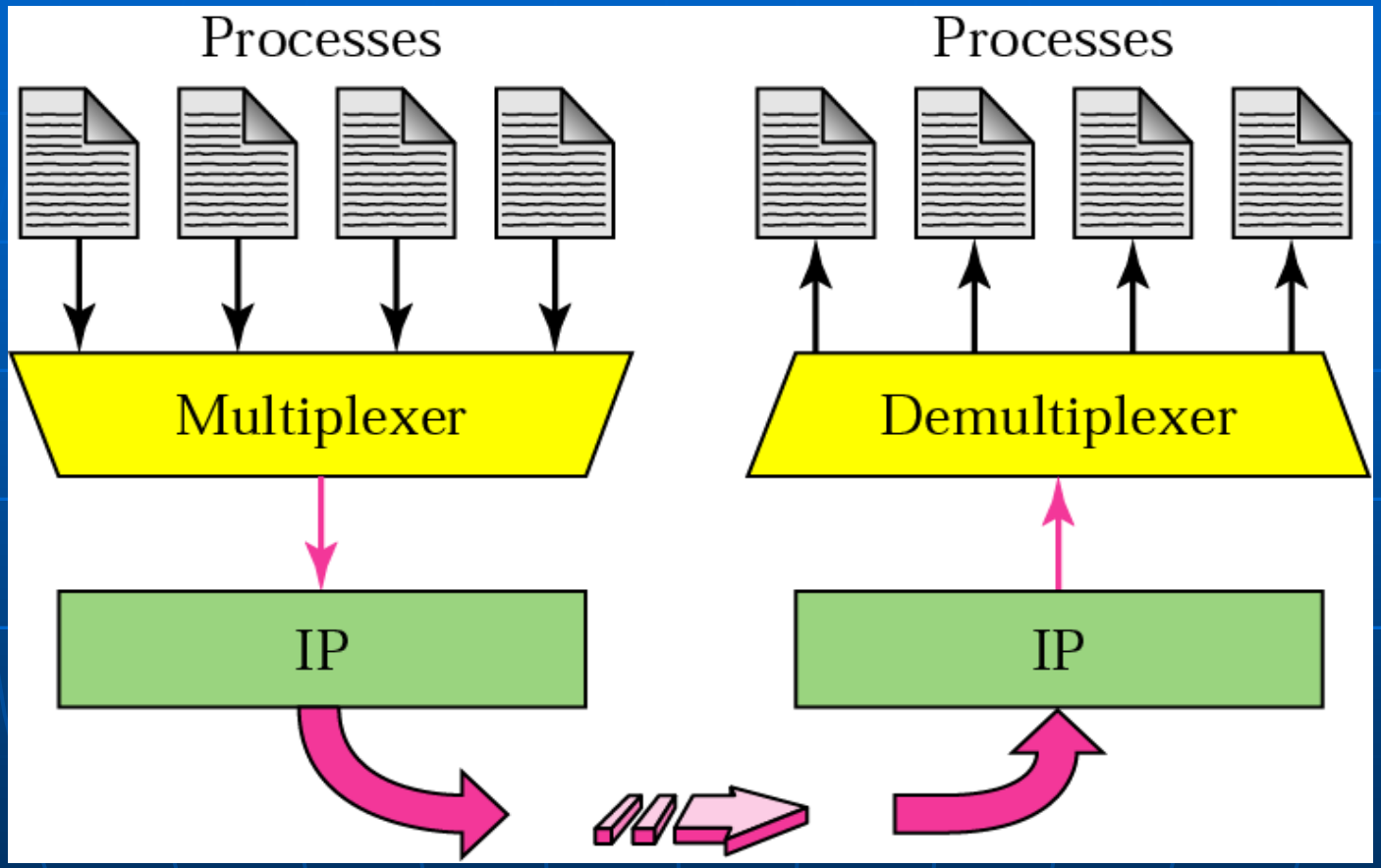


Figure 22.7 Connection establishment

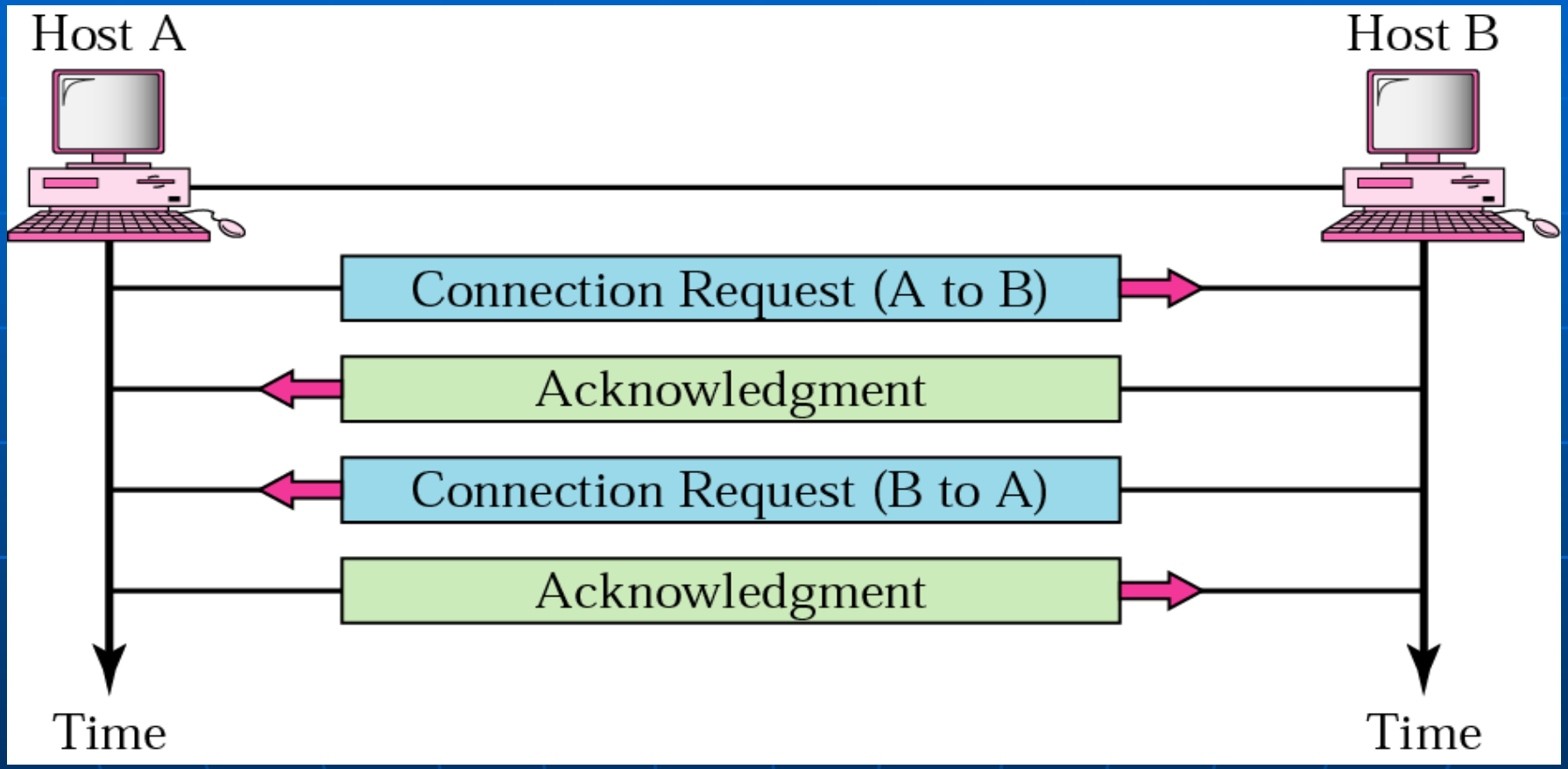


Figure 22.8 Connection termination

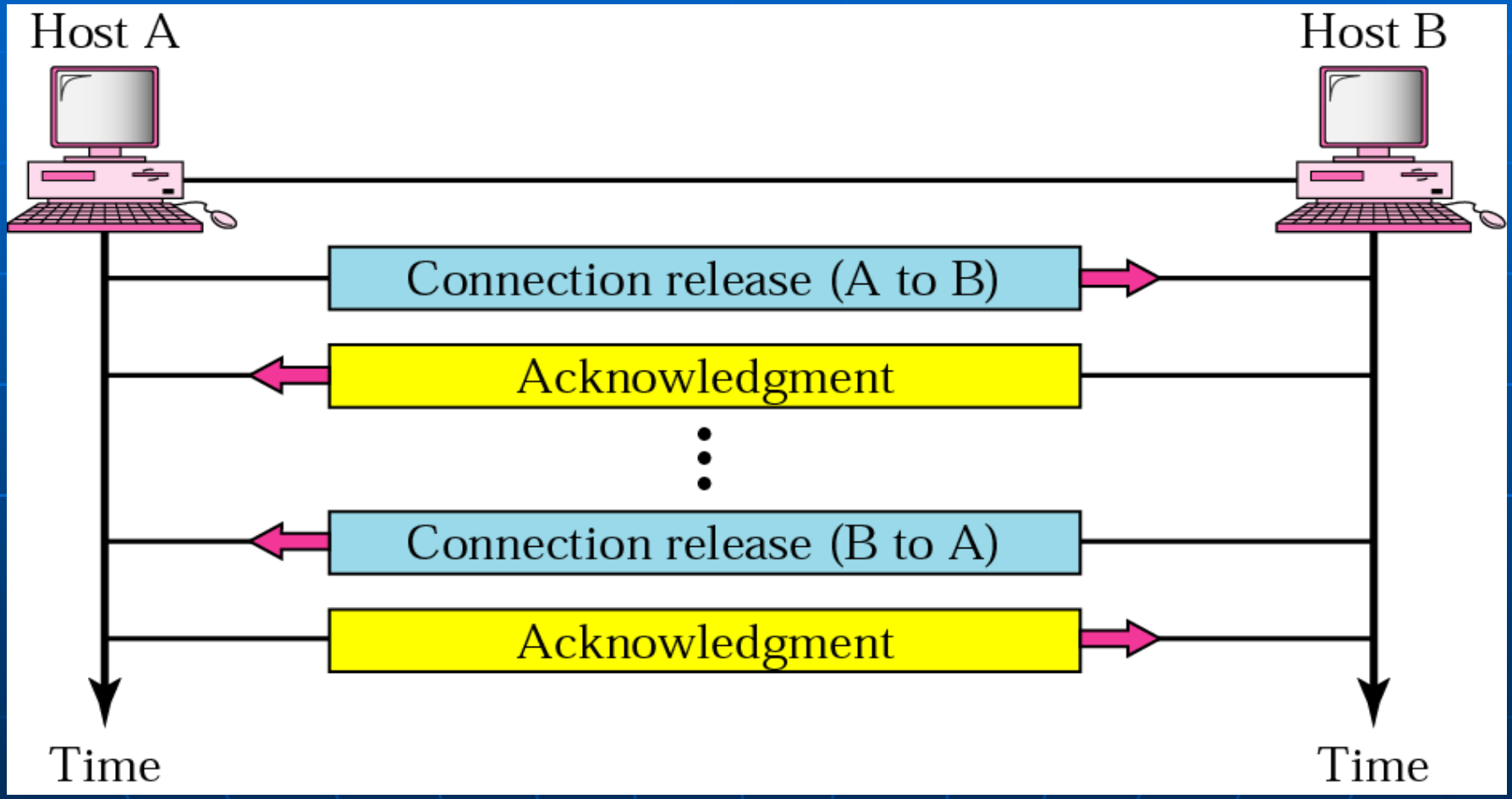
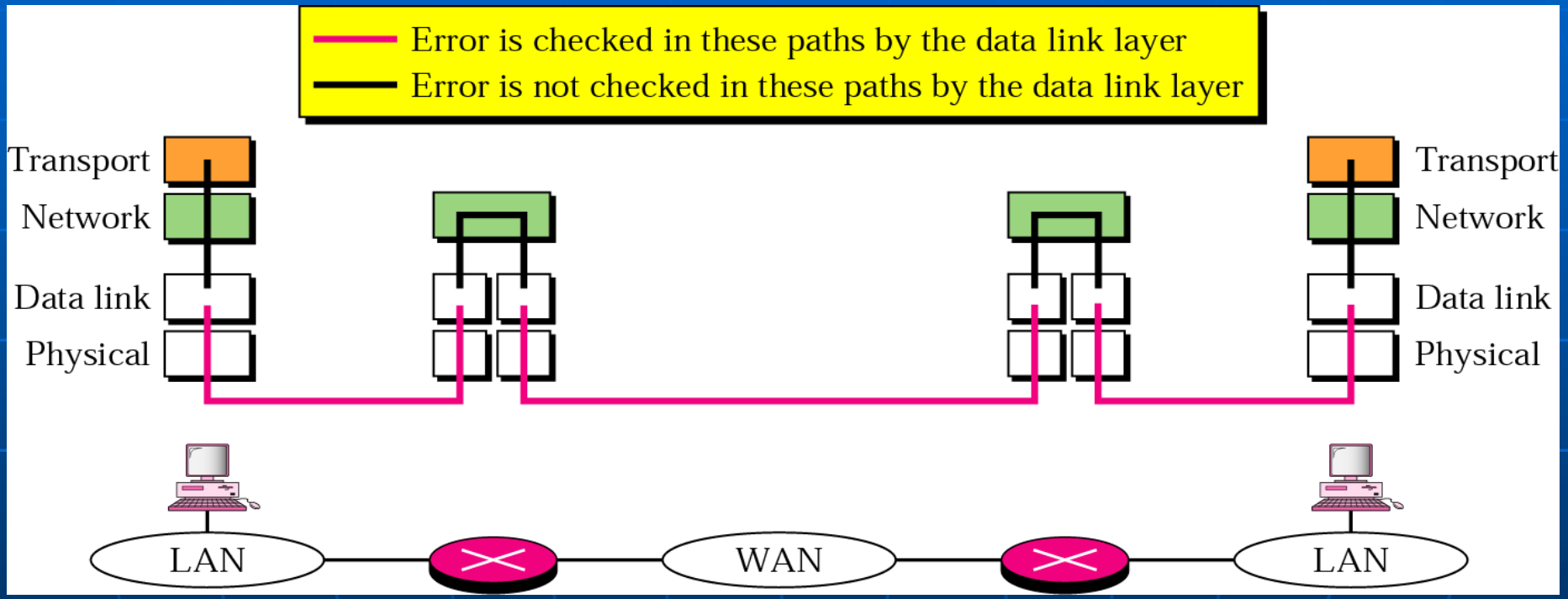


Figure 22.9 Error control



22.2 UDP

Port Numbers

User Datagram

Applications



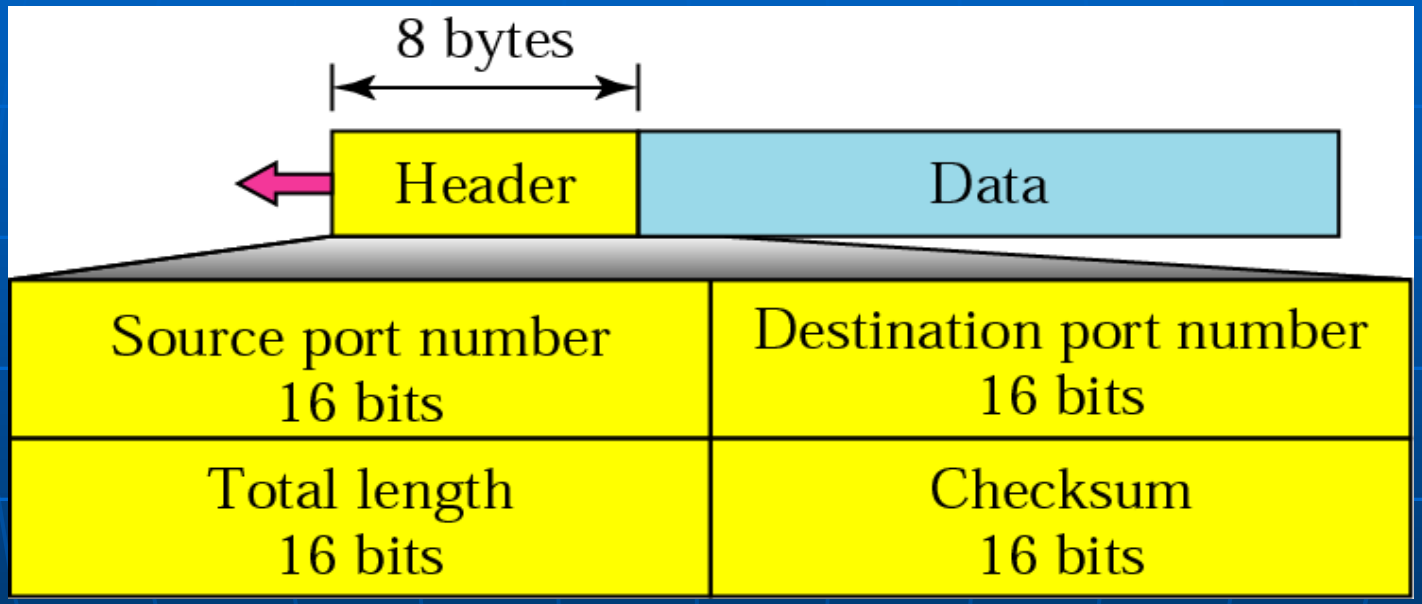
Note:

UDP is a connectionless, unreliable protocol that has no flow and error control. It uses port numbers to multiplex data from the application layer.

Table 22.1 Well-known ports used by UDP

Port	*Protocol*	*Description*
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	Bootps	Server port to download bootstrap information
68	Bootpc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

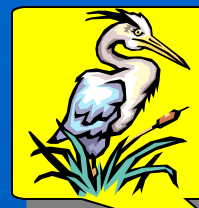
Figure 22.10 User datagram format





Note:

The calculation of checksum and its inclusion in the user datagram are optional.



Note:

UDP is a convenient transport-layer protocol for applications that provide flow and error control. It is also used by multimedia applications.

22.3 TCP

Port Numbers

Services

Sequence Numbers

Segments

Connection

Transition Diagram

Flow and Error Control

Silly Window Syndrome

Table 22.2 Well-known ports used by TCP

Port	Protocol	Description
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

Figure 22.11 **Stream delivery**

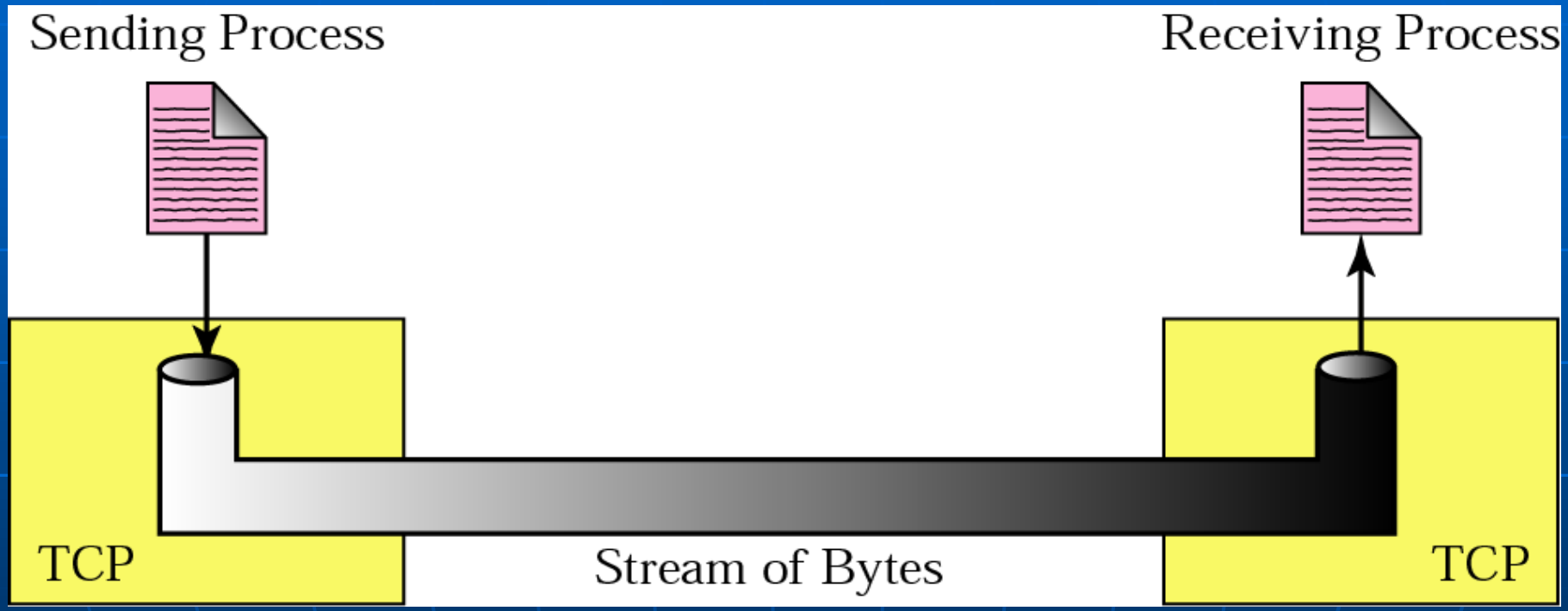


Figure 22.12 Sending and receiving buffers

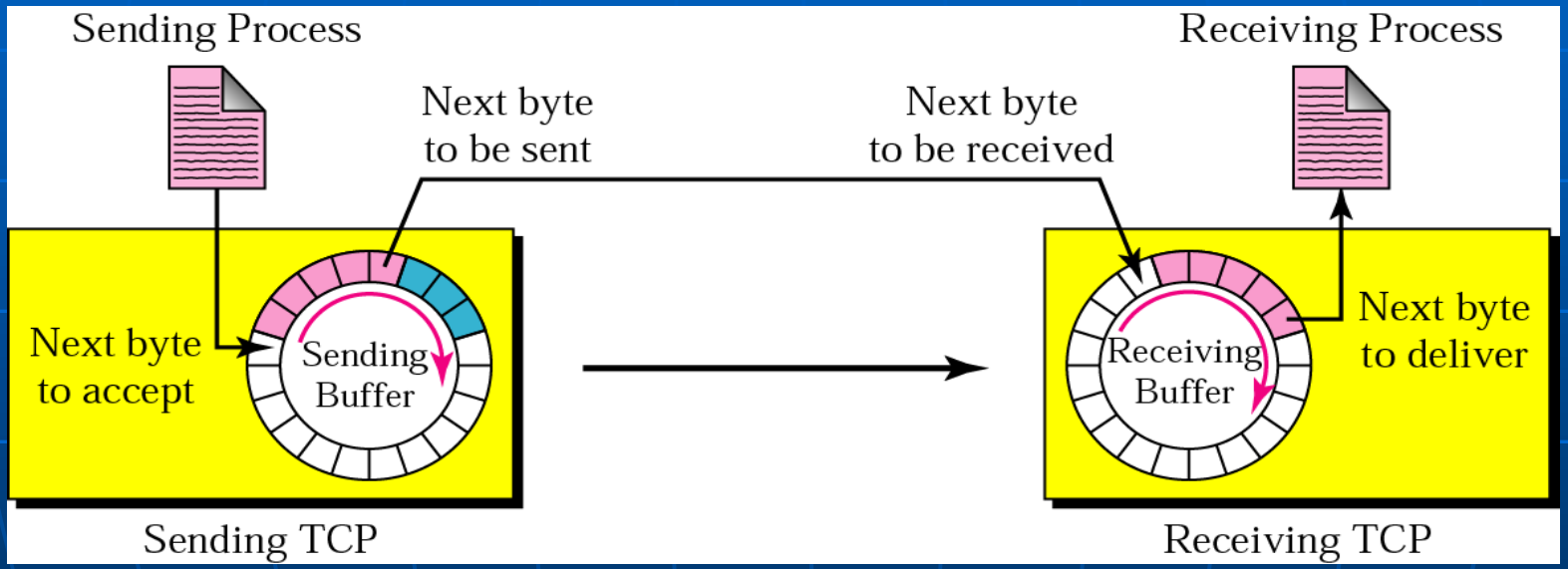
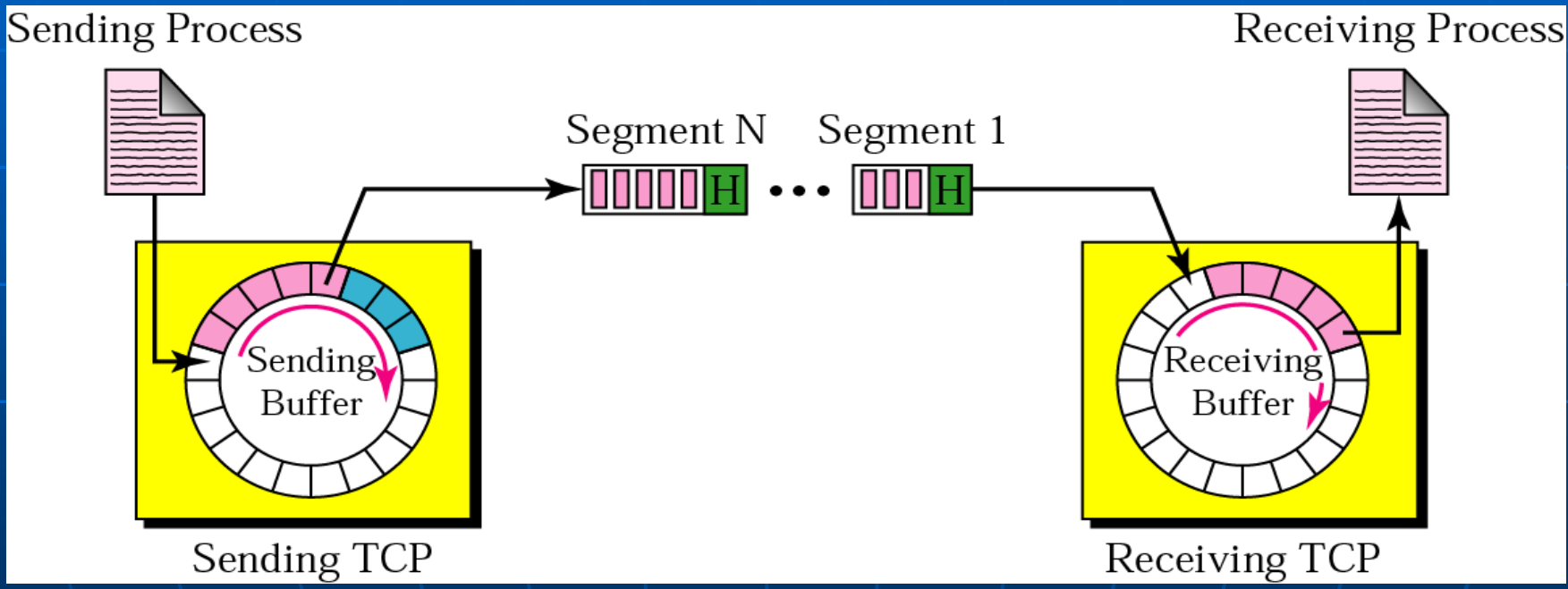


Figure 22.13 TCP segments



Example 1

Imagine a TCP connection is transferring a file of 6000 bytes. The first byte is numbered 10010. What are the sequence numbers for each segment if data are sent in five segments with the first four segments carrying 1000 bytes and the last segment carrying 2000 bytes?

Solution

The following shows the sequence number for each segment:

Segment 1 ==> sequence number: 10,010 (range: 10,010 to 11,009)
Segment 2 ==> sequence number: 11,010 (range: 11,010 to 12,009)
Segment 3 ==> sequence number: 12,010 (range: 12,010 to 13,009)
Segment 4 ==> sequence number: 13,010 (range: 13,010 to 14,009)
Segment 5 ==> sequence number: 14,010 (range: 14,010 to 16,009)



Note:

The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.



Note:

The value of the sequence number field in a segment defines the number of the first data byte contained in that segment.



Note:

The value of the acknowledgment field in a segment defines the number of the next byte a party expects to receive. The acknowledgment number is cumulative.

Figure 22.14 TCP segment format

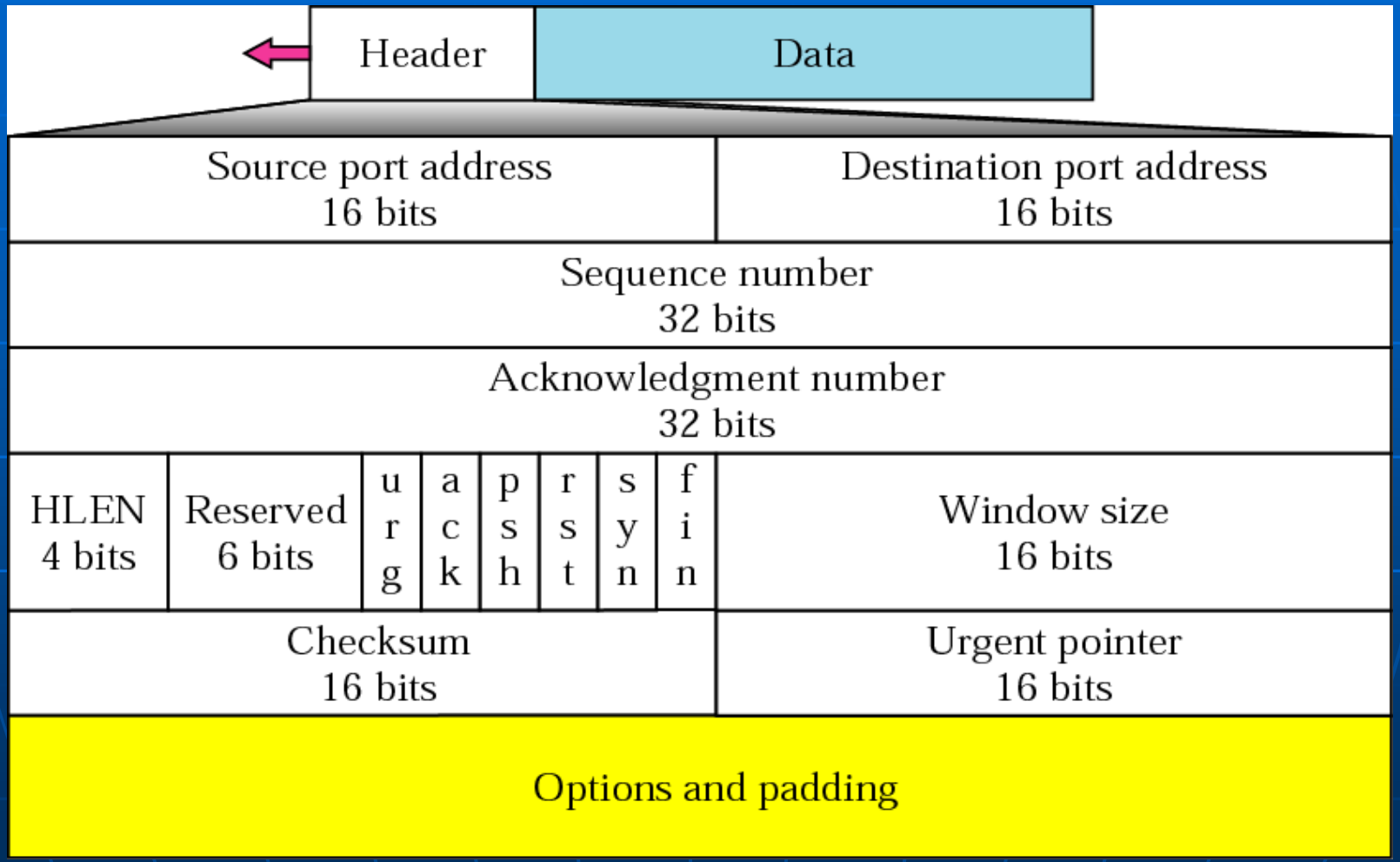


Figure 22.15 Control field

URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection



Table 22.3 Description of flags in the control field

Flag	Description
URG	The value of the urgent pointer field is valid.
ACK	The value of the acknowledgment field is valid.
PSH	Push the data.
RST	The connection must be reset.
SYN	Synchronize sequence numbers during connection.
FIN	Terminate the connection.

Figure 22.16 Three-step connection establishment

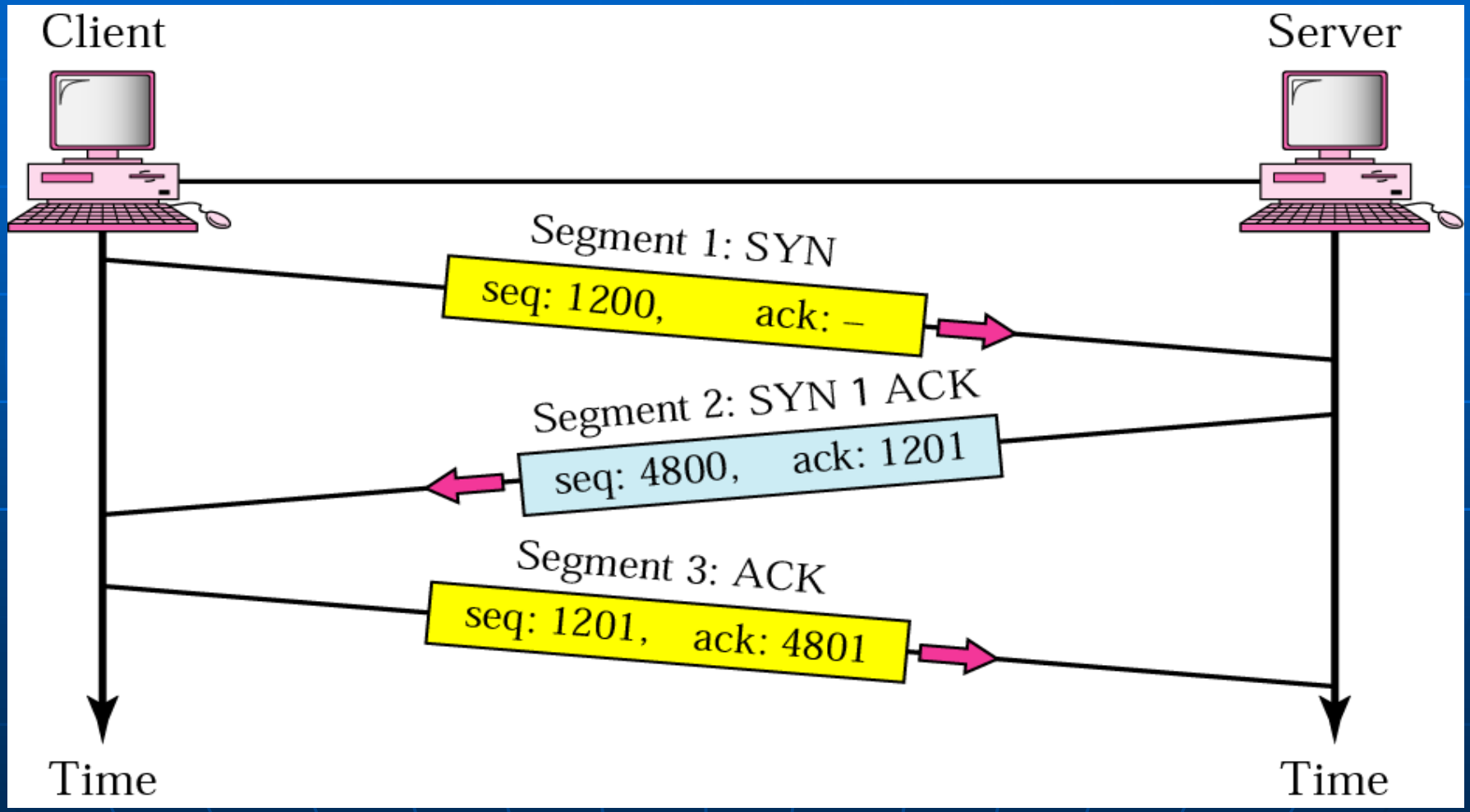


Figure 22.17 Four-step connection termination

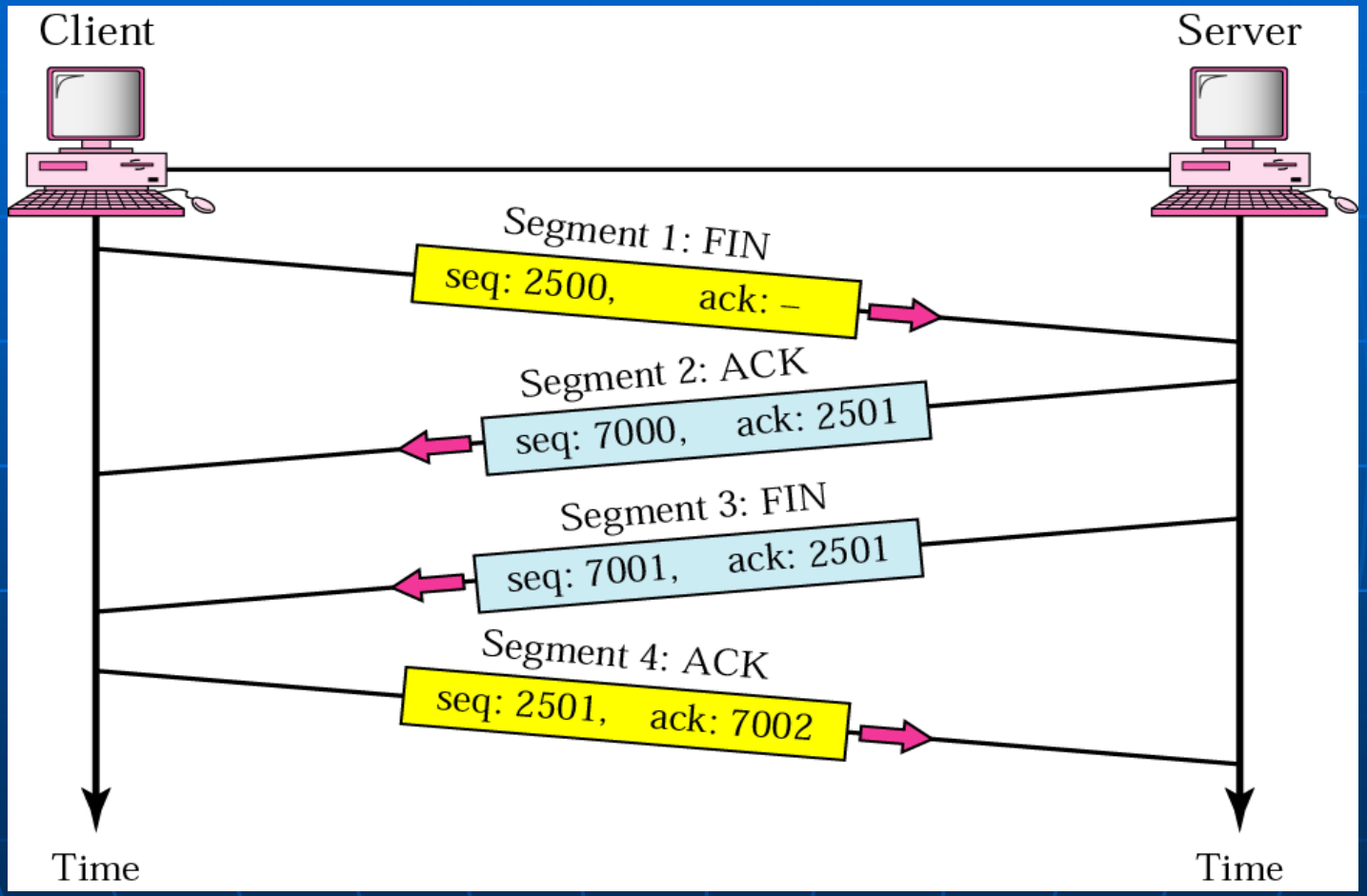
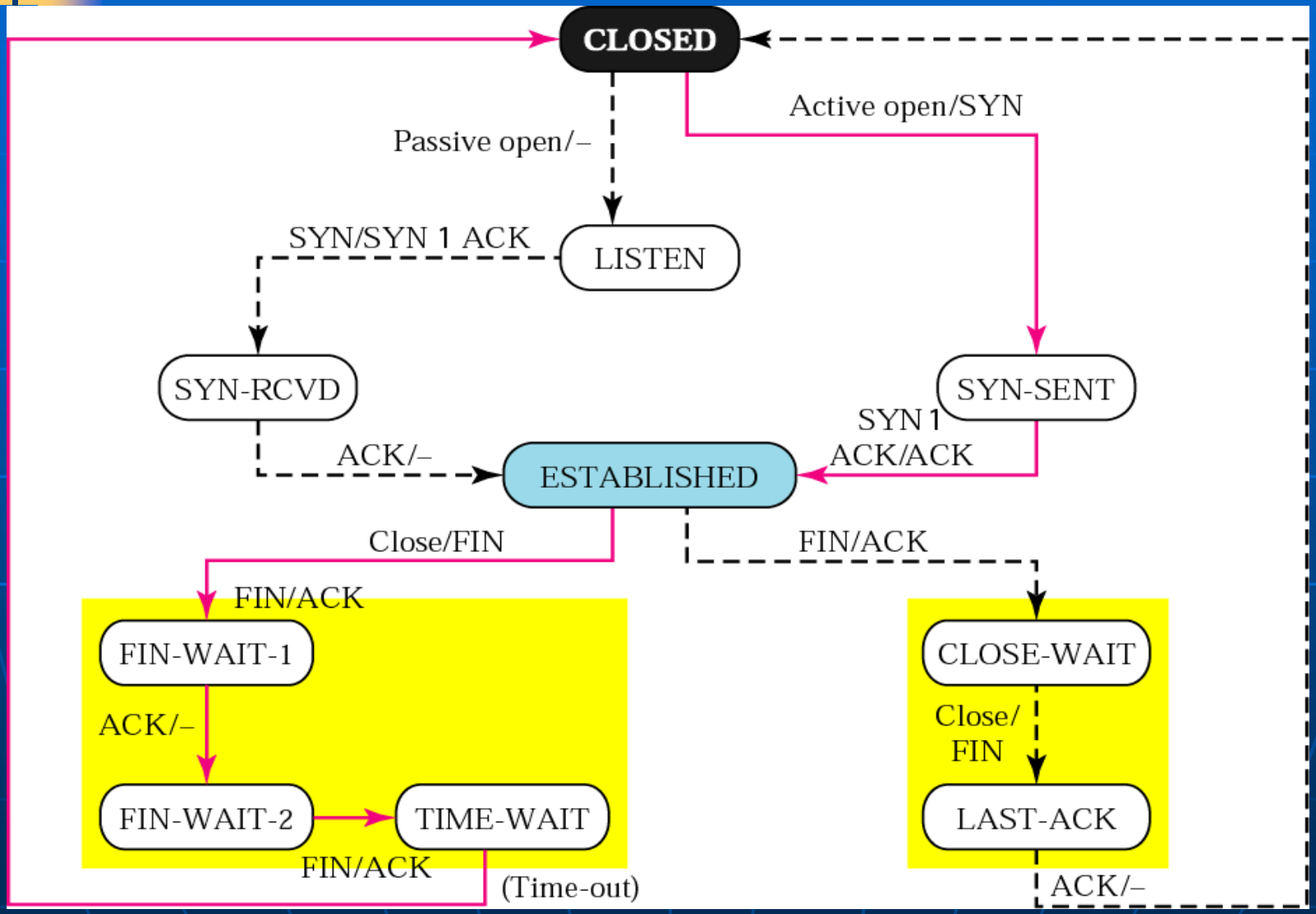


Table 22.4 States for TCP

State	Description
CLOSED	There is no connection.
LISTEN	The server is waiting for calls from the client.
SYN-SENT	A connection request is sent; waiting for acknowledgment.
SYN-RCVD	A connection request is received.
ESTABLISHED	Connection is established.
FIN-WAIT-1	The application has requested the closing of the connection.
FIN-WAIT-2	The other side has accepted the closing of the connection.
TIME-WAIT	Waiting for retransmitted segments to die.
CLOSE-WAIT	The server is waiting for the application to close.
LAST-ACK	The server is waiting for the last acknowledgment.

Figure 22.18 State transition diagram





Note:

A sliding window is used to make transmission more efficient as well as to control the flow of data so that the destination does not become overwhelmed with data. TCP's sliding windows are byte-oriented.

Figure 22.19 Sender buffer

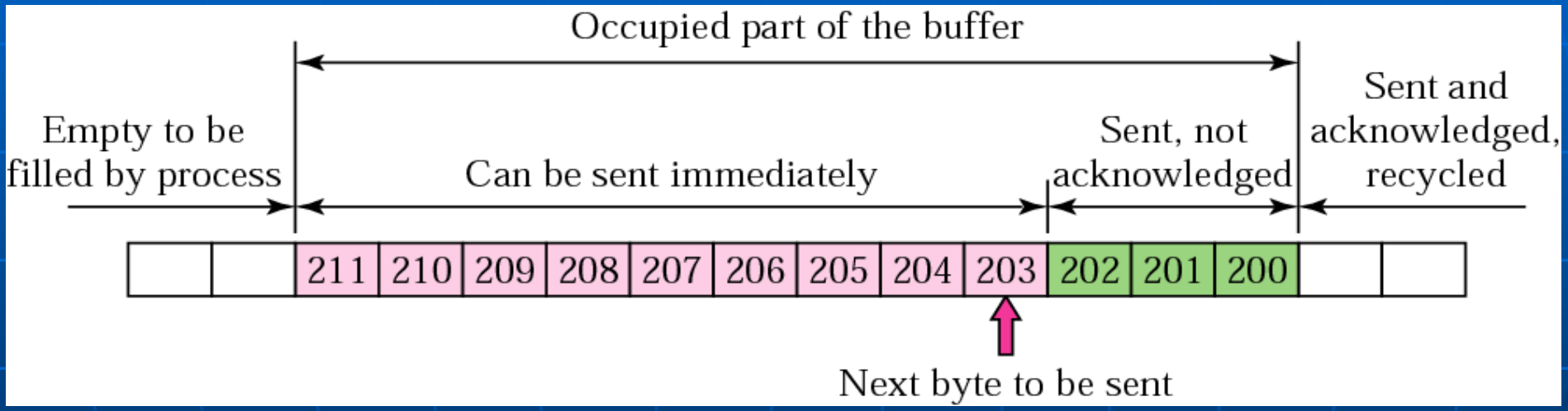


Figure 22.20 Receiver window

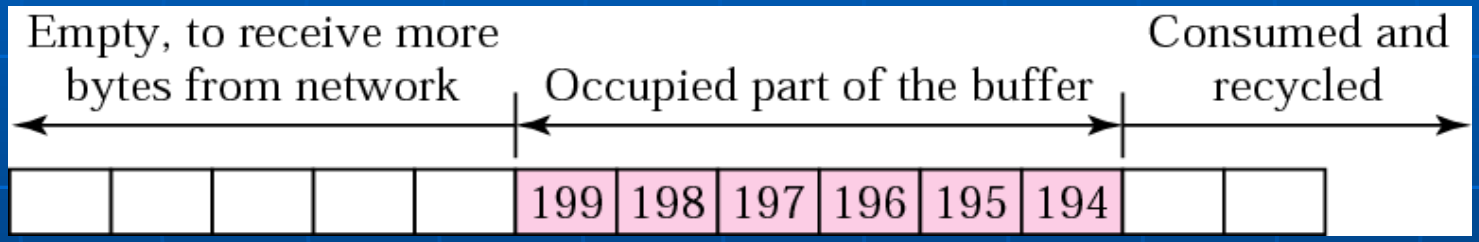


Figure 22.21 Sender buffer and sender window

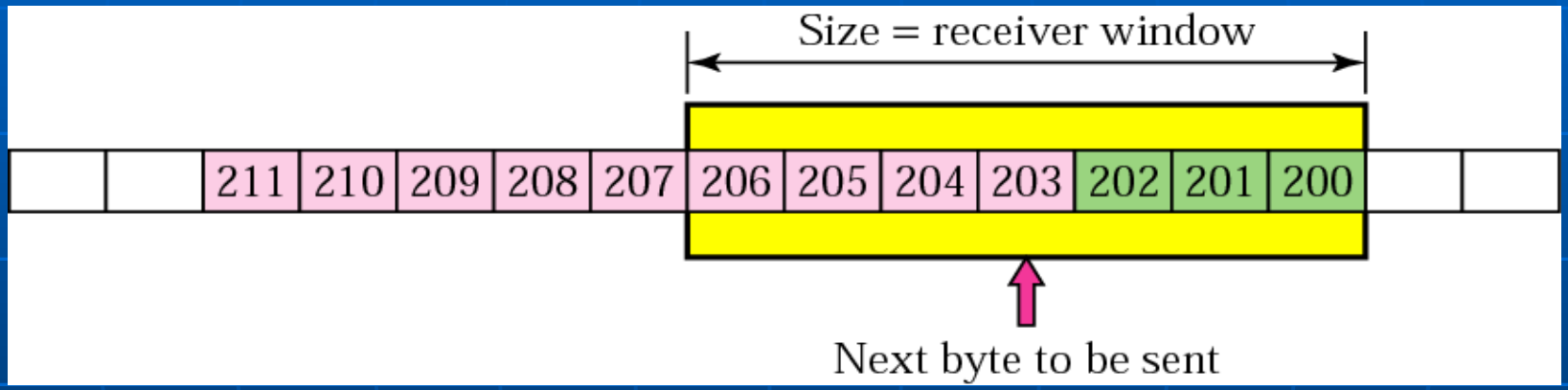


Figure 22.22 Sliding the sender window

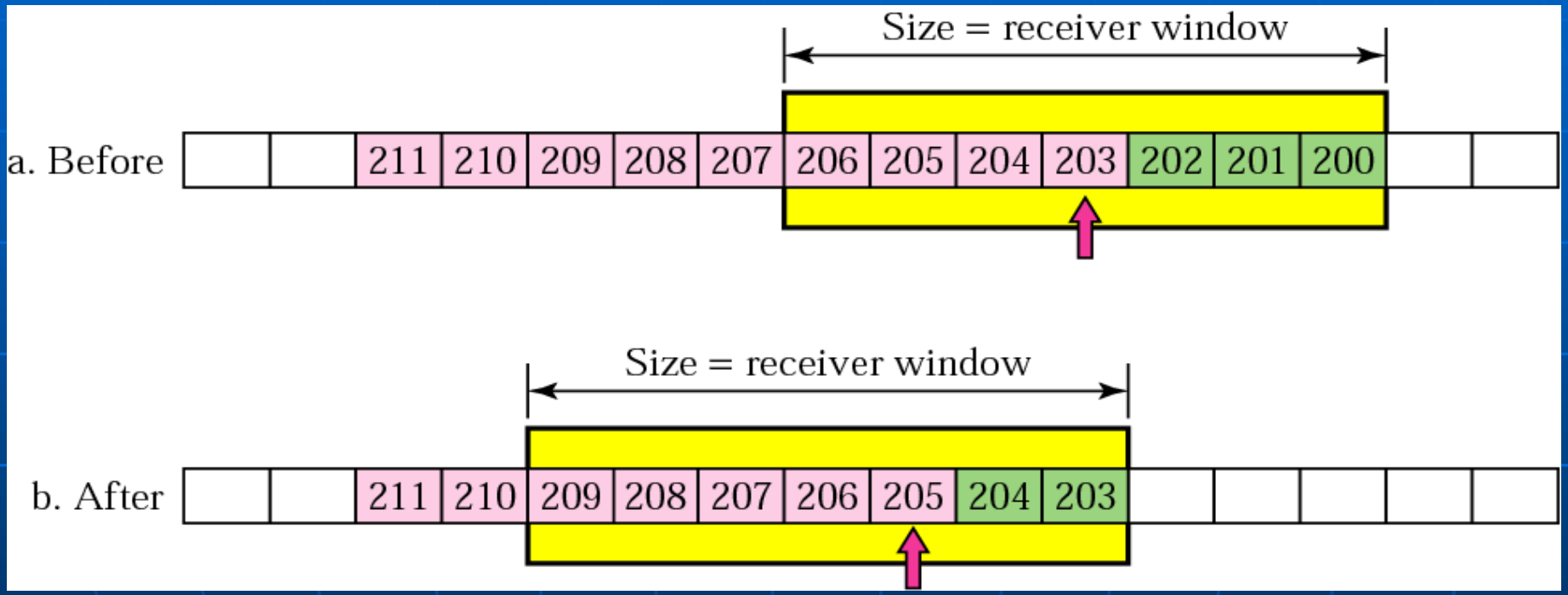


Figure 22.23 Expanding the sender window

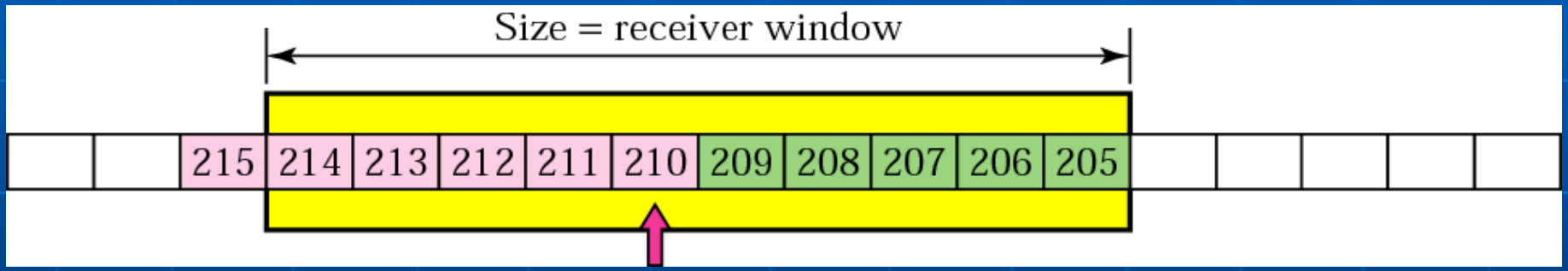
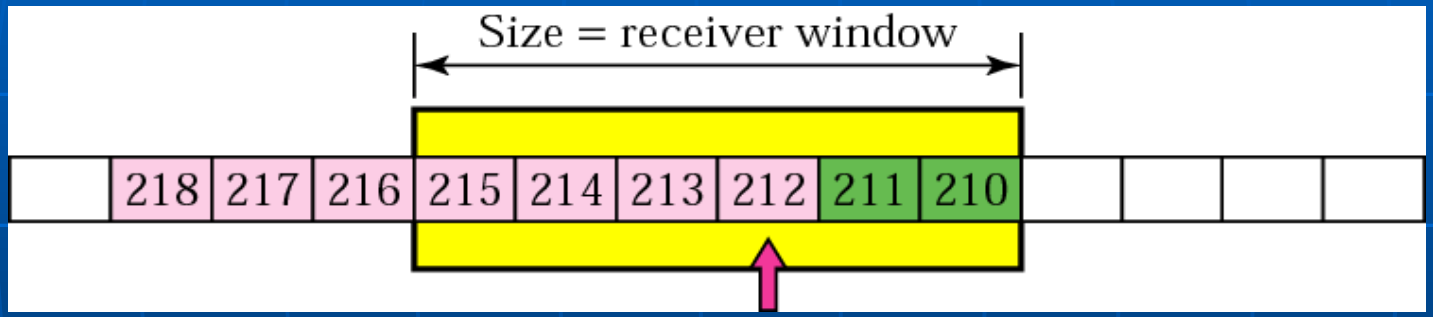


Figure 22.24 Shrinking the sender window





Note:

In TCP, the sender window size is totally controlled by the receiver window value (the number of empty locations in the receiver buffer). However, the actual window size can be smaller if there is congestion in the network.



Note:

Some points about TCP's sliding windows:

- **The source does not have to send a full window's worth of data.**
- **The size of the window can be increased or decreased by the destination.**
- **The destination can send an acknowledgment at any time.**

Figure 22.25 Lost segment

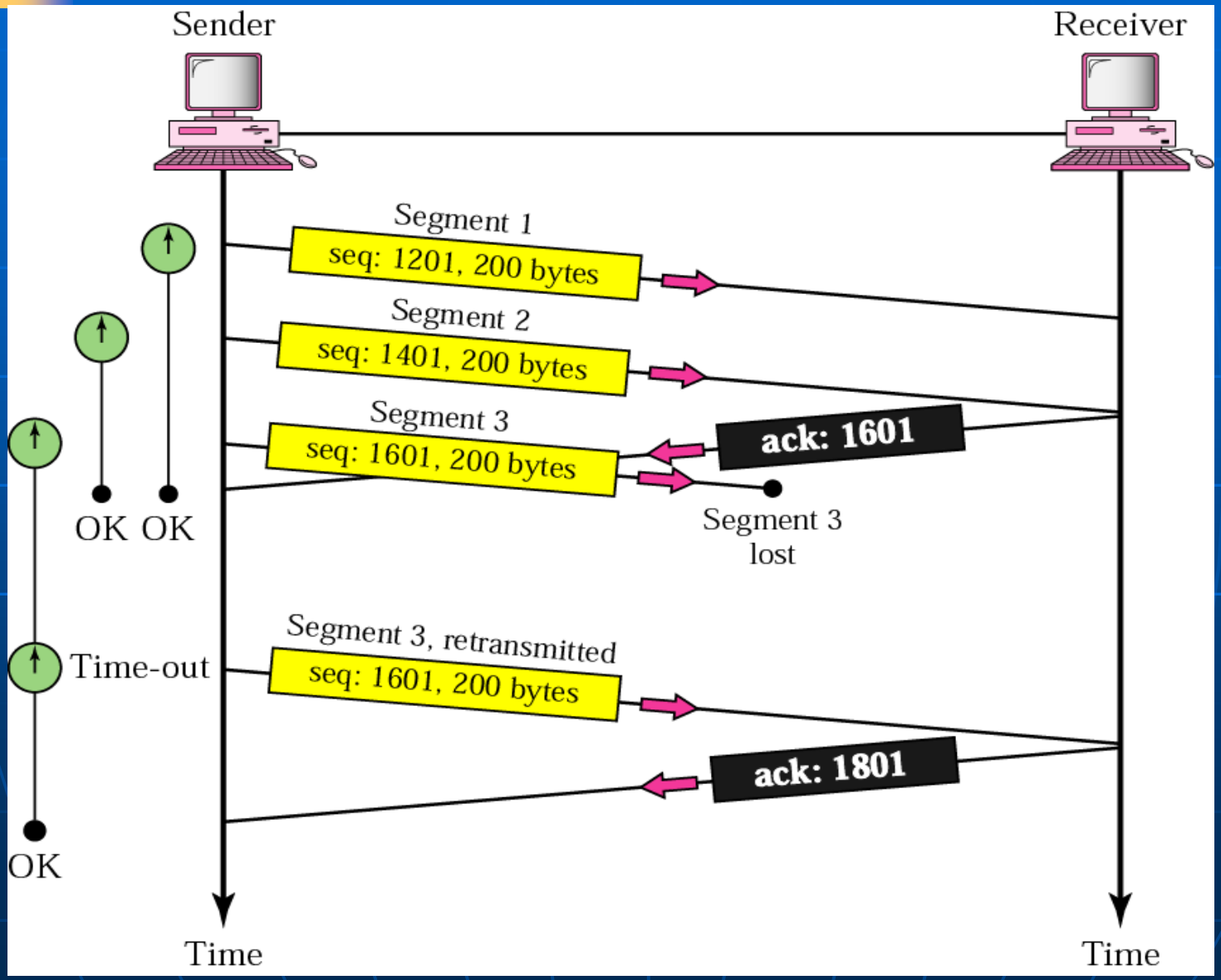


Figure 22.26 Lost acknowledgment

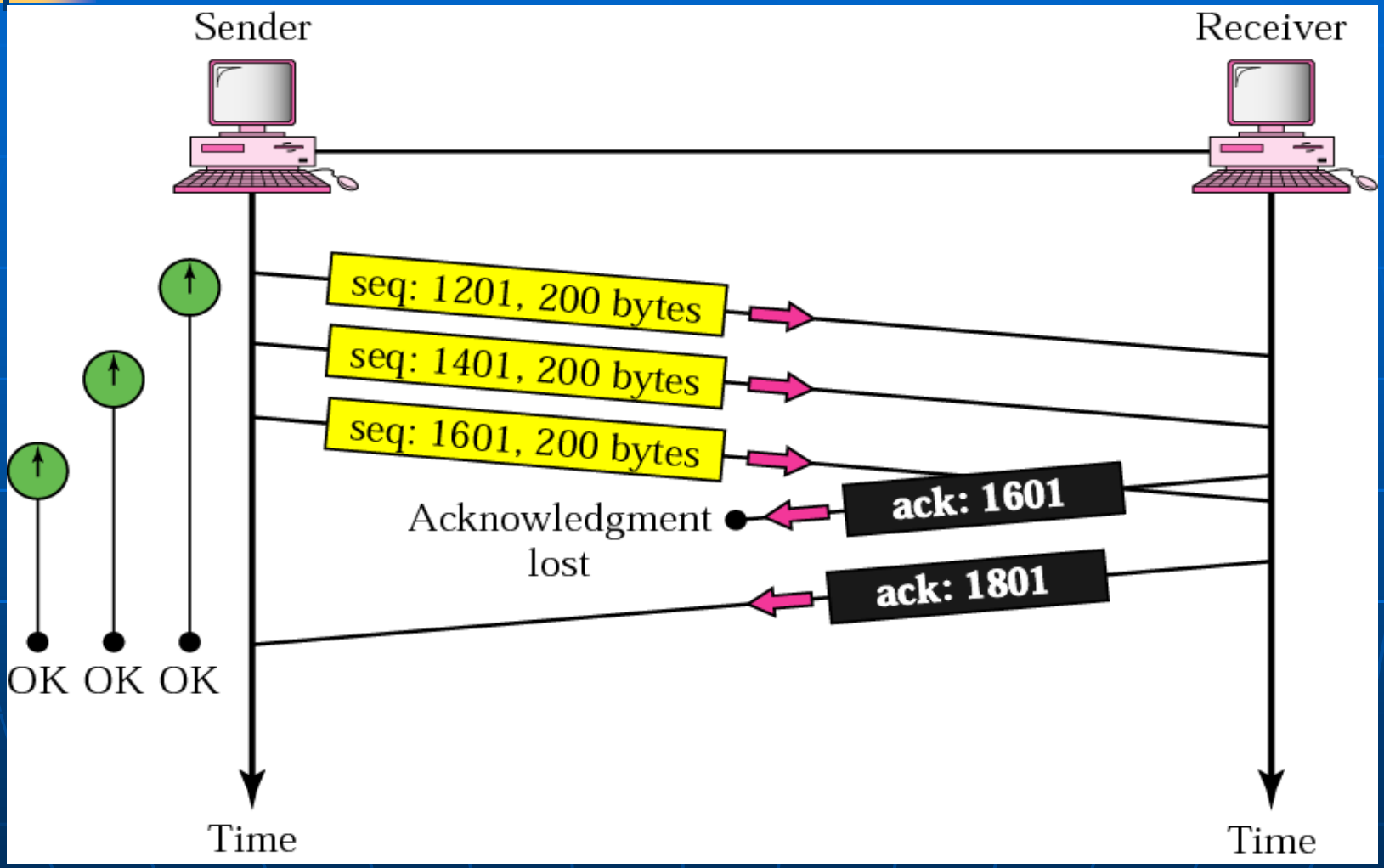
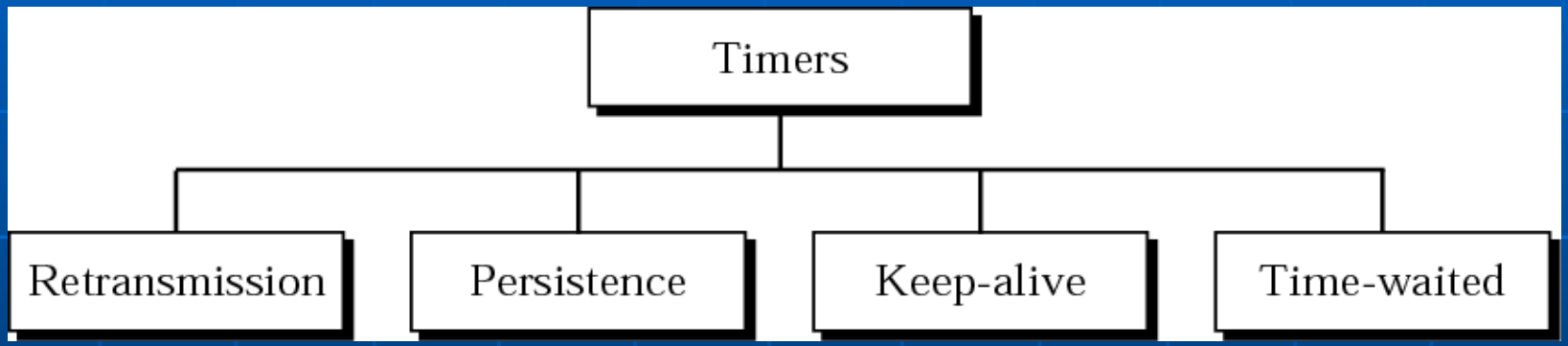


Figure 22.27 TCP timers



Chapter 23

Congestion Control and Quality of Service

23.1 Data Traffic

Traffic Descriptor

Traffic Profiles

Figure 23.1 Traffic descriptors

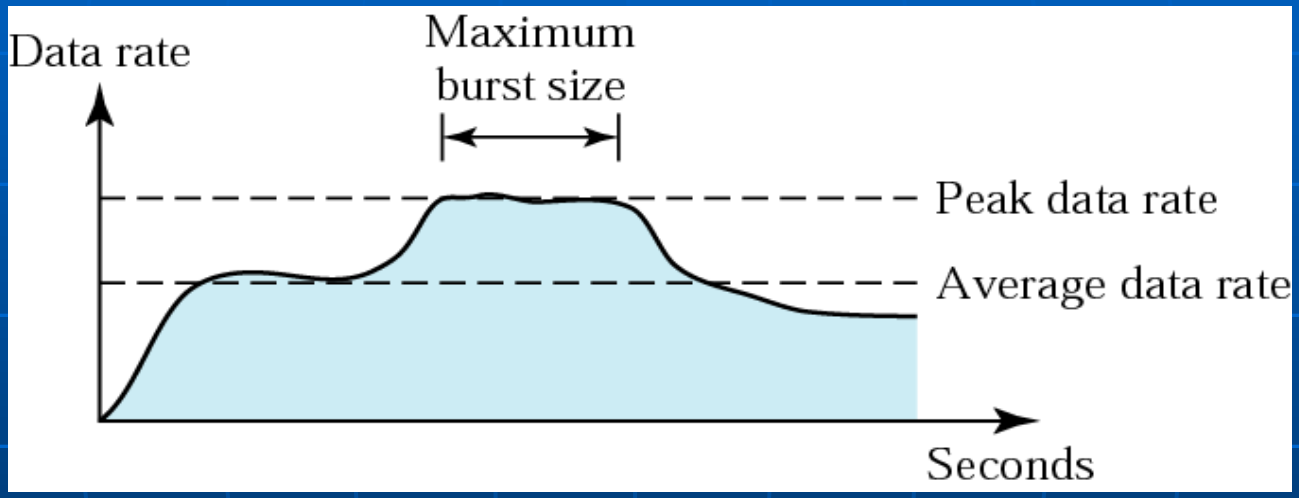


Figure 23.2 Constant-bit-rate traffic



Figure 23.3 Variable-bit-rate traffic

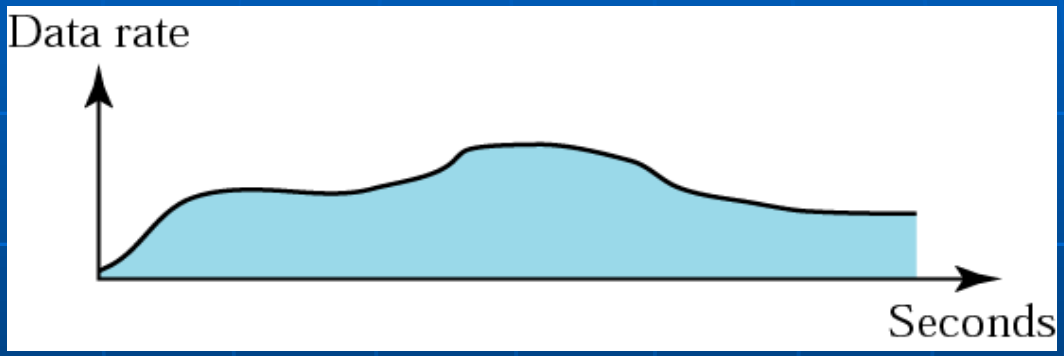
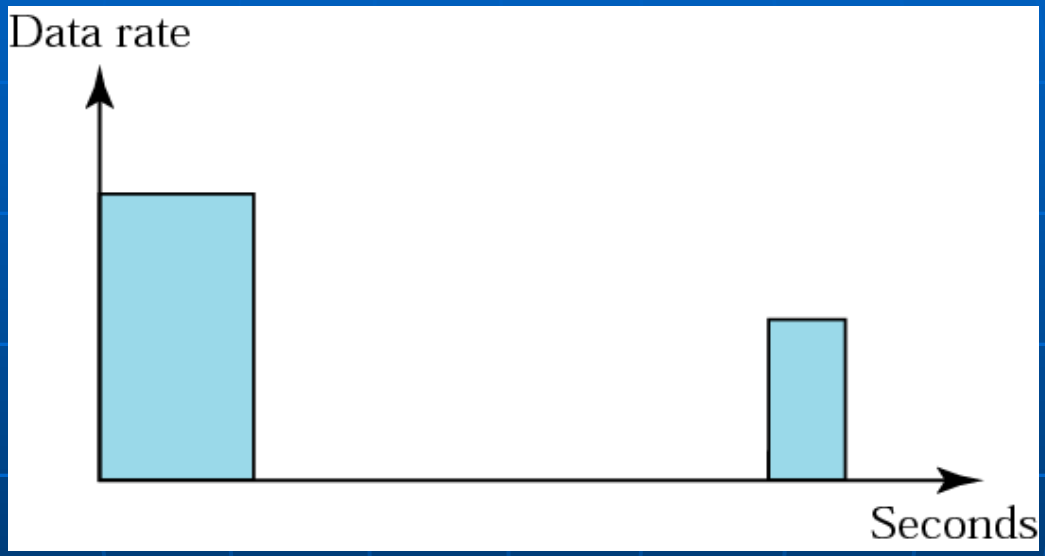


Figure 23.4 Bursty traffic



23.2 Congestion

Network Performance

Figure 23.5 Incoming packet

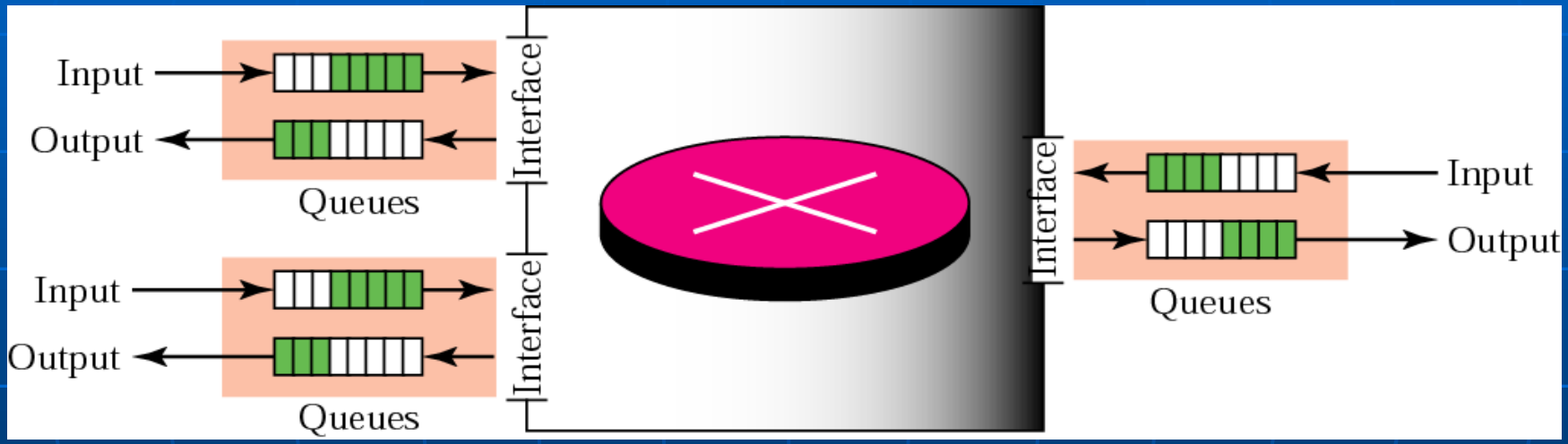


Figure 23.6 Packet delay and network load

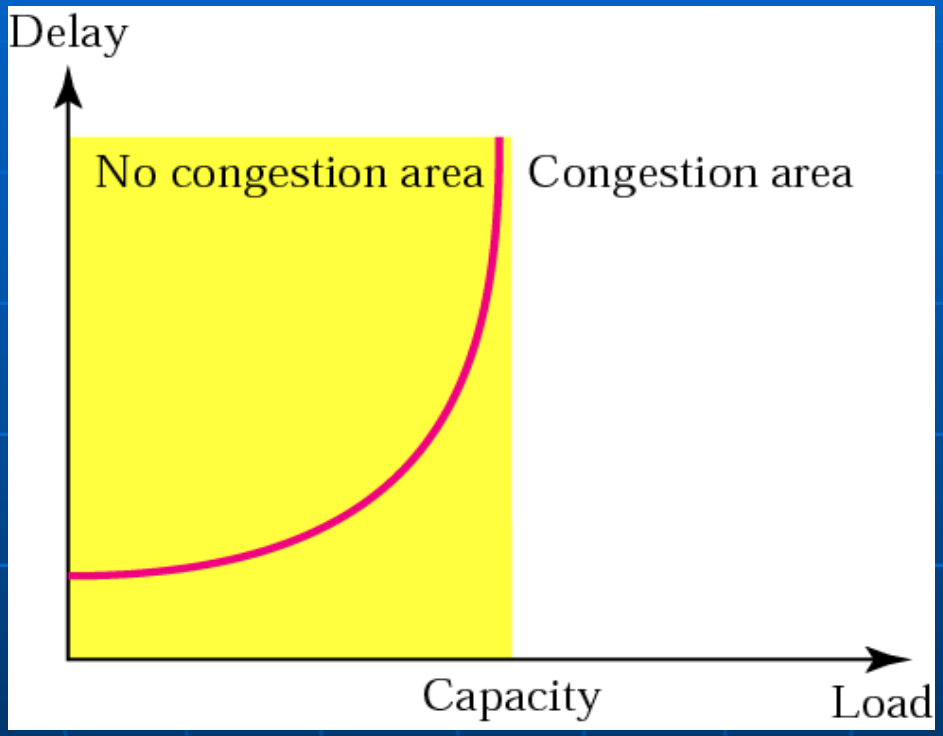
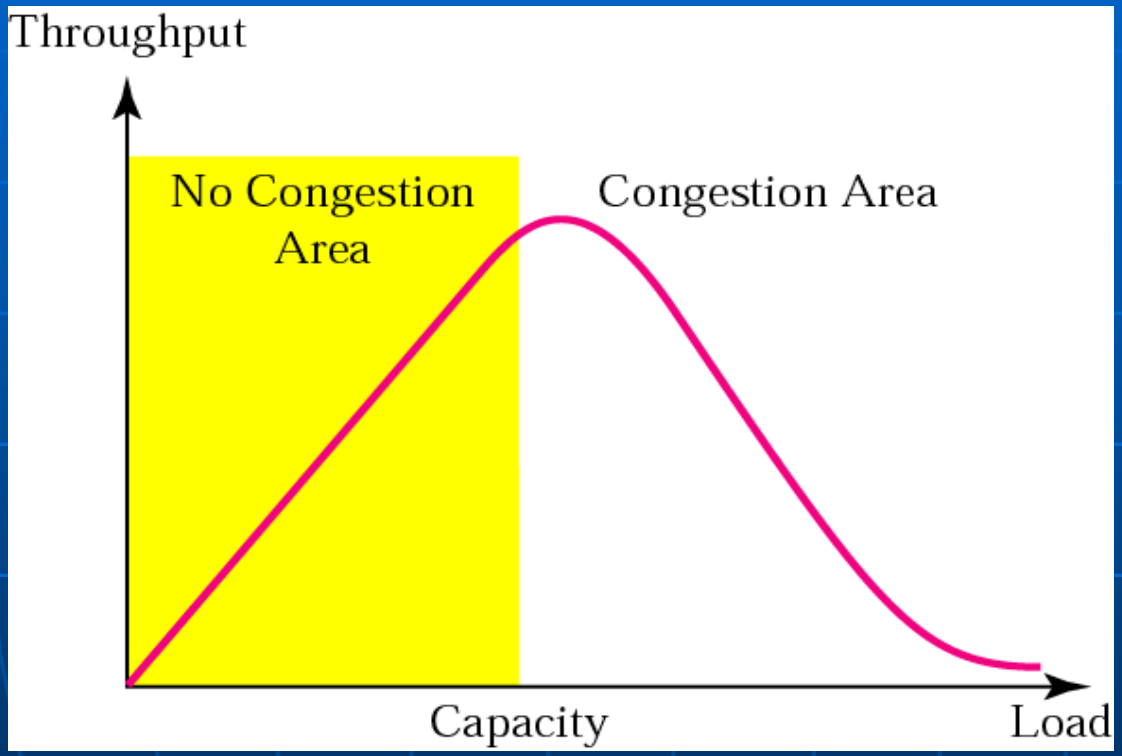


Figure 23.7 Throughput versus network load



23.3 Congestion Control

Open Loop

Closed Loop

23.4 Two Examples

Congestion Control in TCP

Congestion Control in Frame Relay



TCP assumes that the cause of a lost segment is due to congestion in the network.



Note:

If the cause of the lost segment is congestion, retransmission of the segment does not remove the cause—it aggravates it.

Figure 23.8 Multiplicative decrease

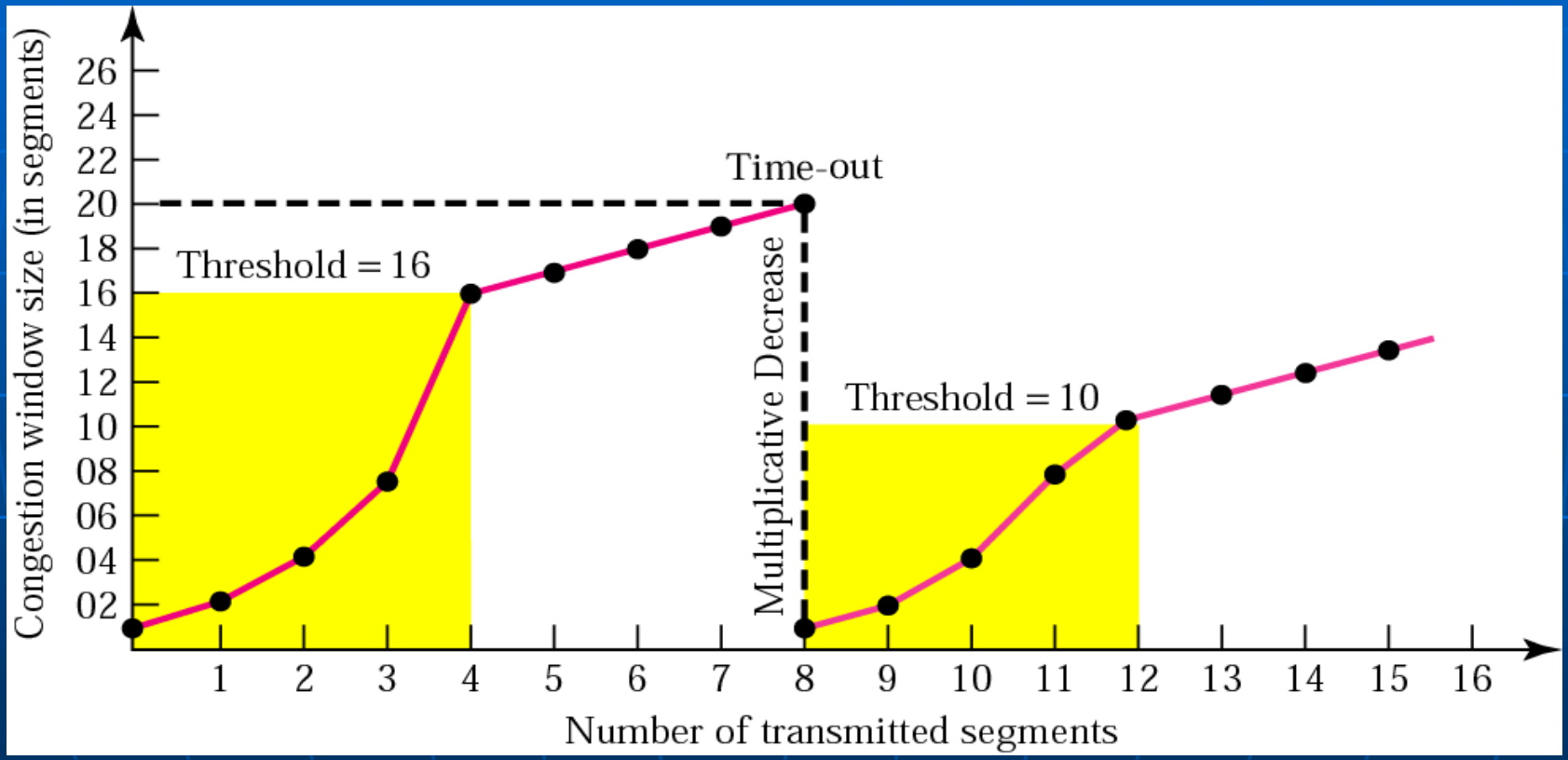


Figure 23.9 BECN

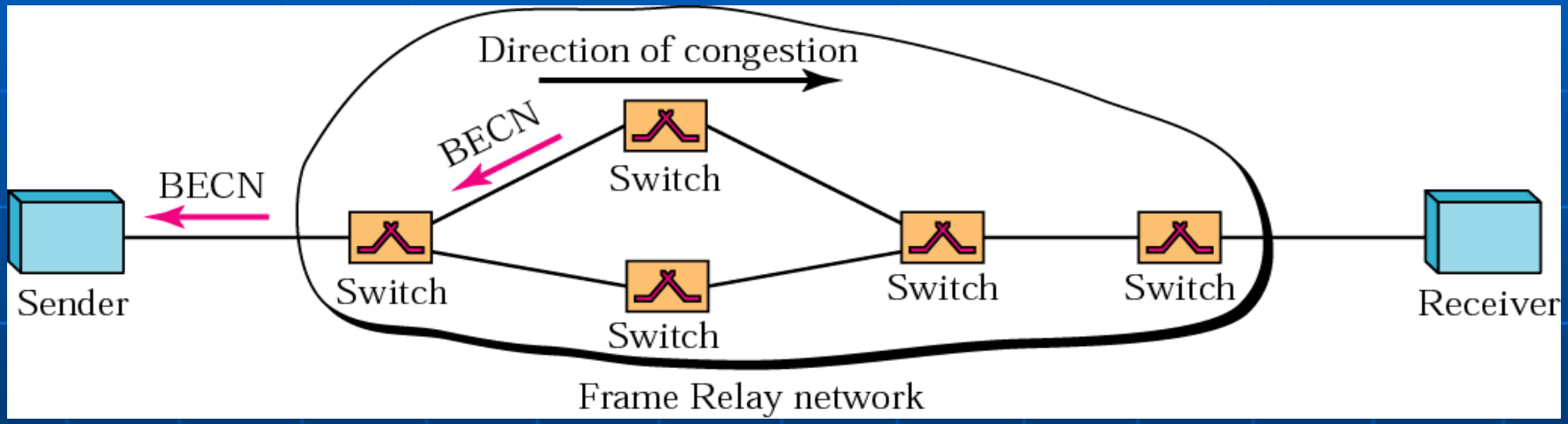


Figure 23.10 FECN

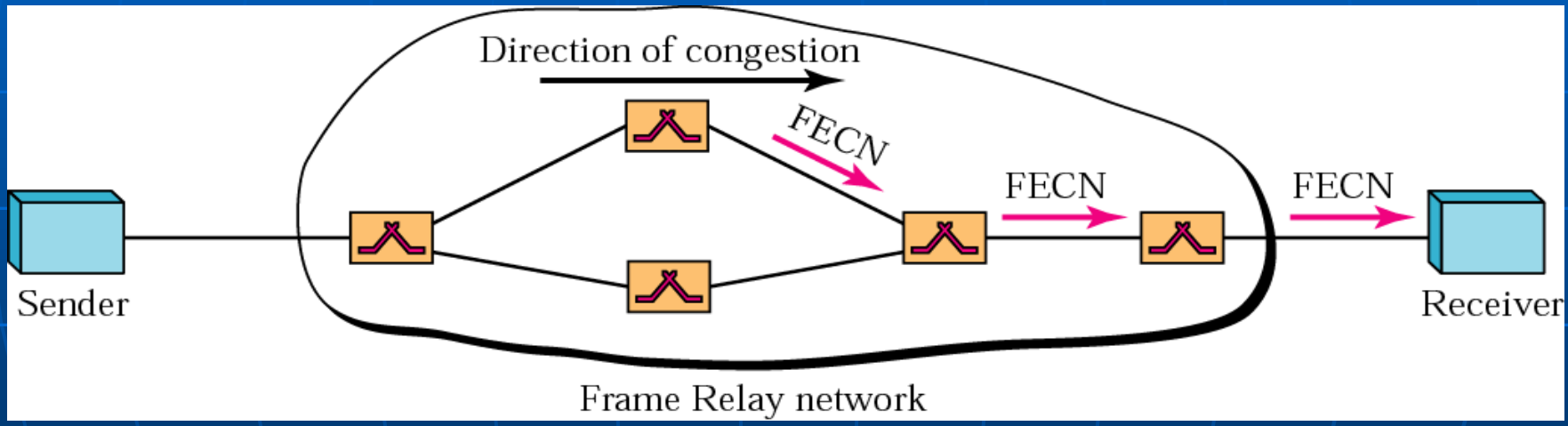
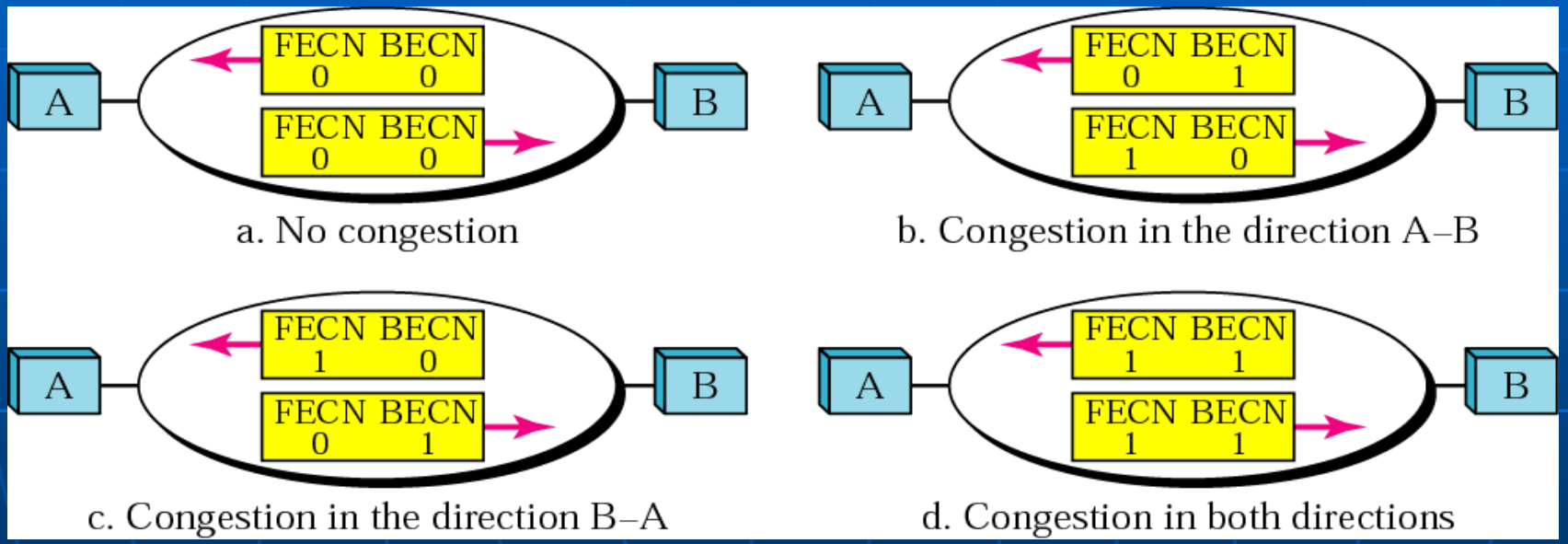


Figure 23.11 Four cases of congestion



23.5 Quality of Service

Flow Characteristics

Flow Classes

23.6 Techniques to Improve QoS

Scheduling

Traffic Shaping

Resource Reservation

Admission Control

Figure 23.12 Flow characteristics

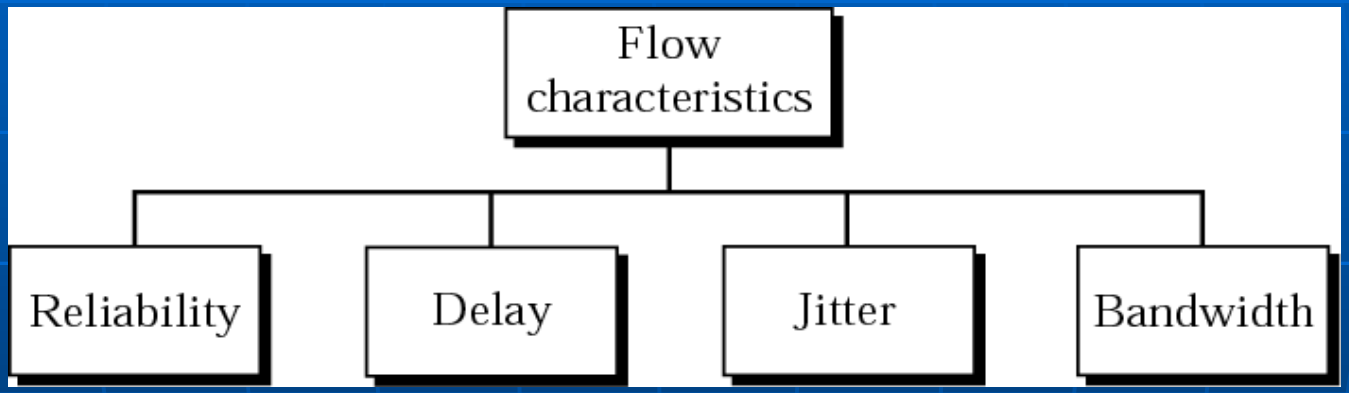


Figure 23.13 FIFO queue

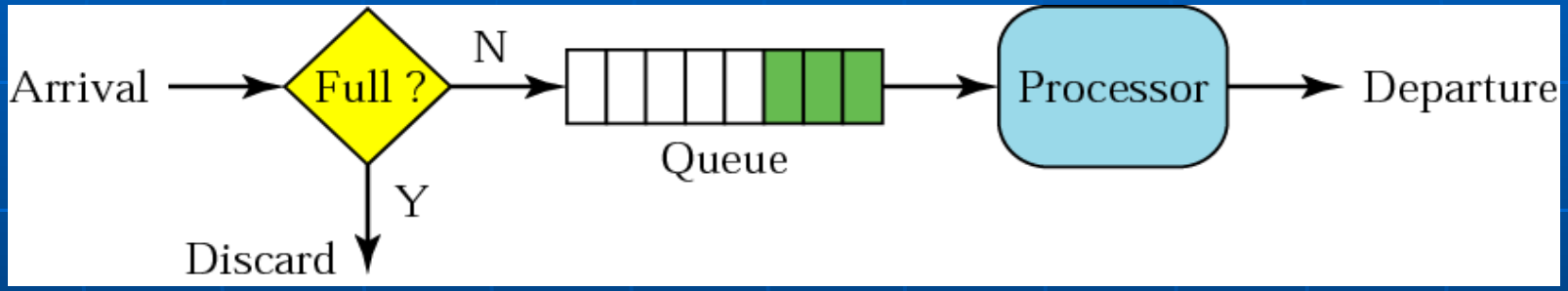


Figure 23.14 Priority queuing

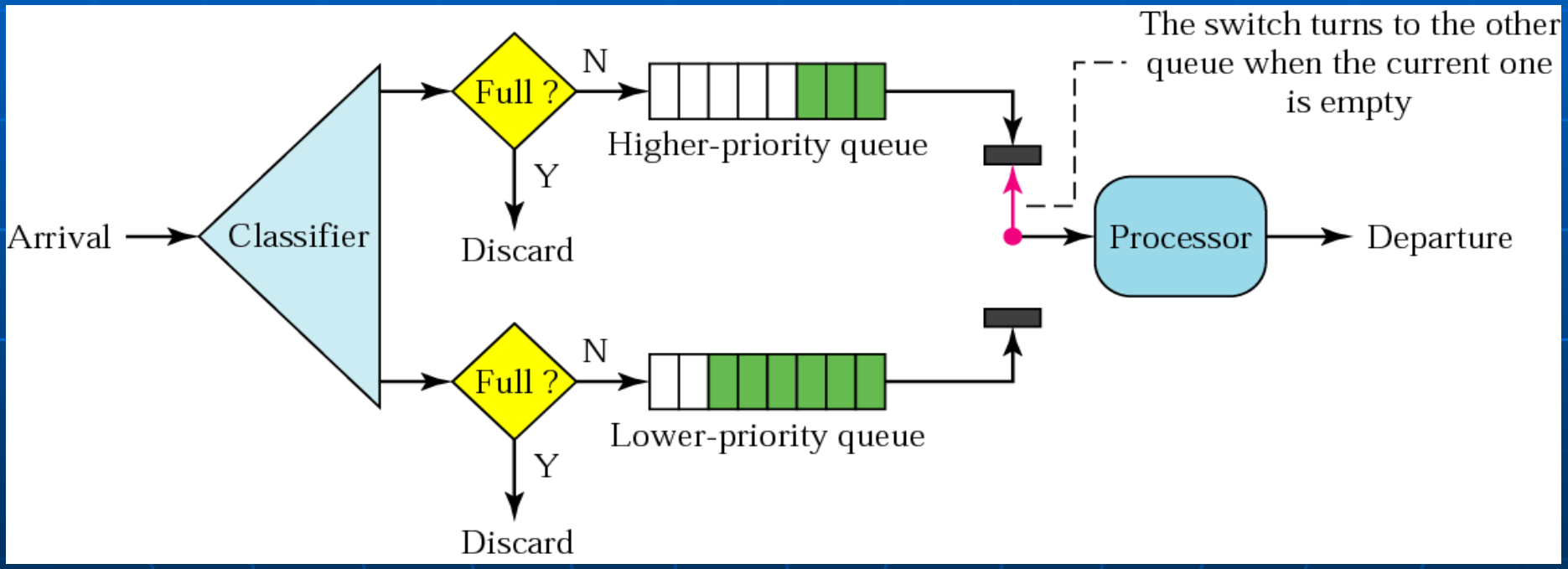


Figure 23.15 Weighted fair queuing

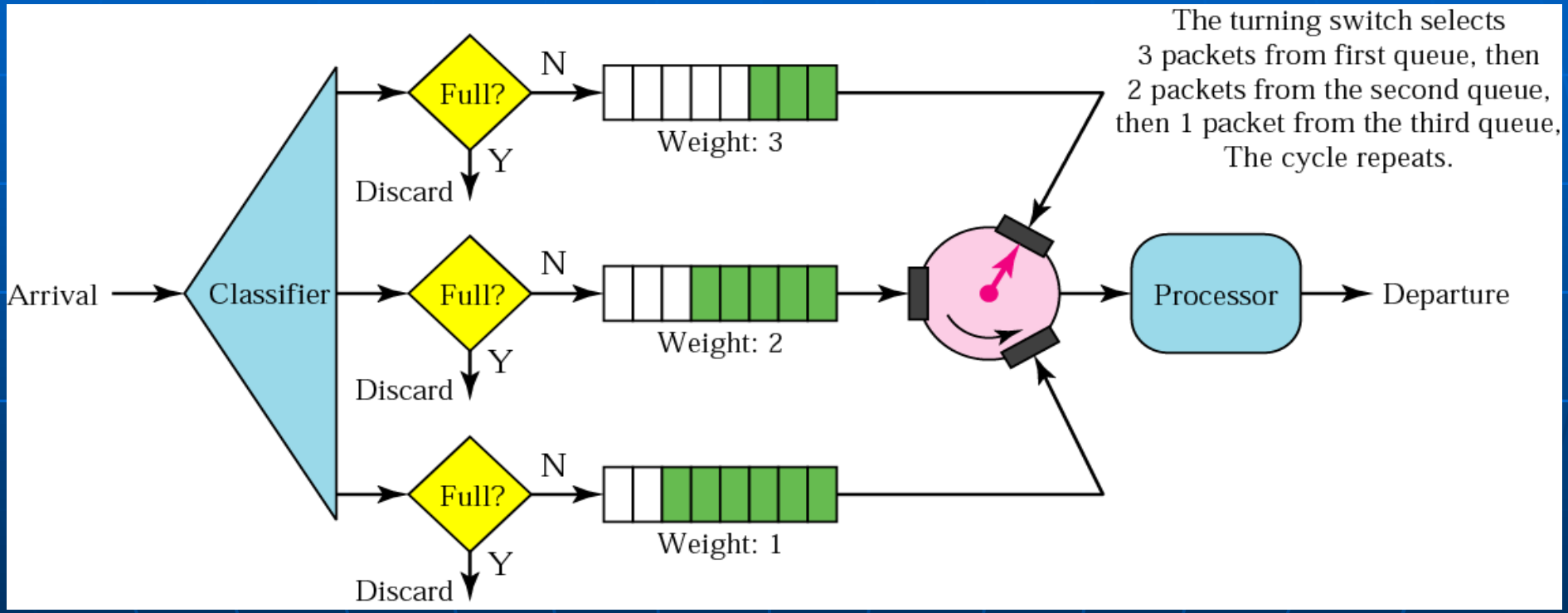


Figure 23.16 Leaky bucket

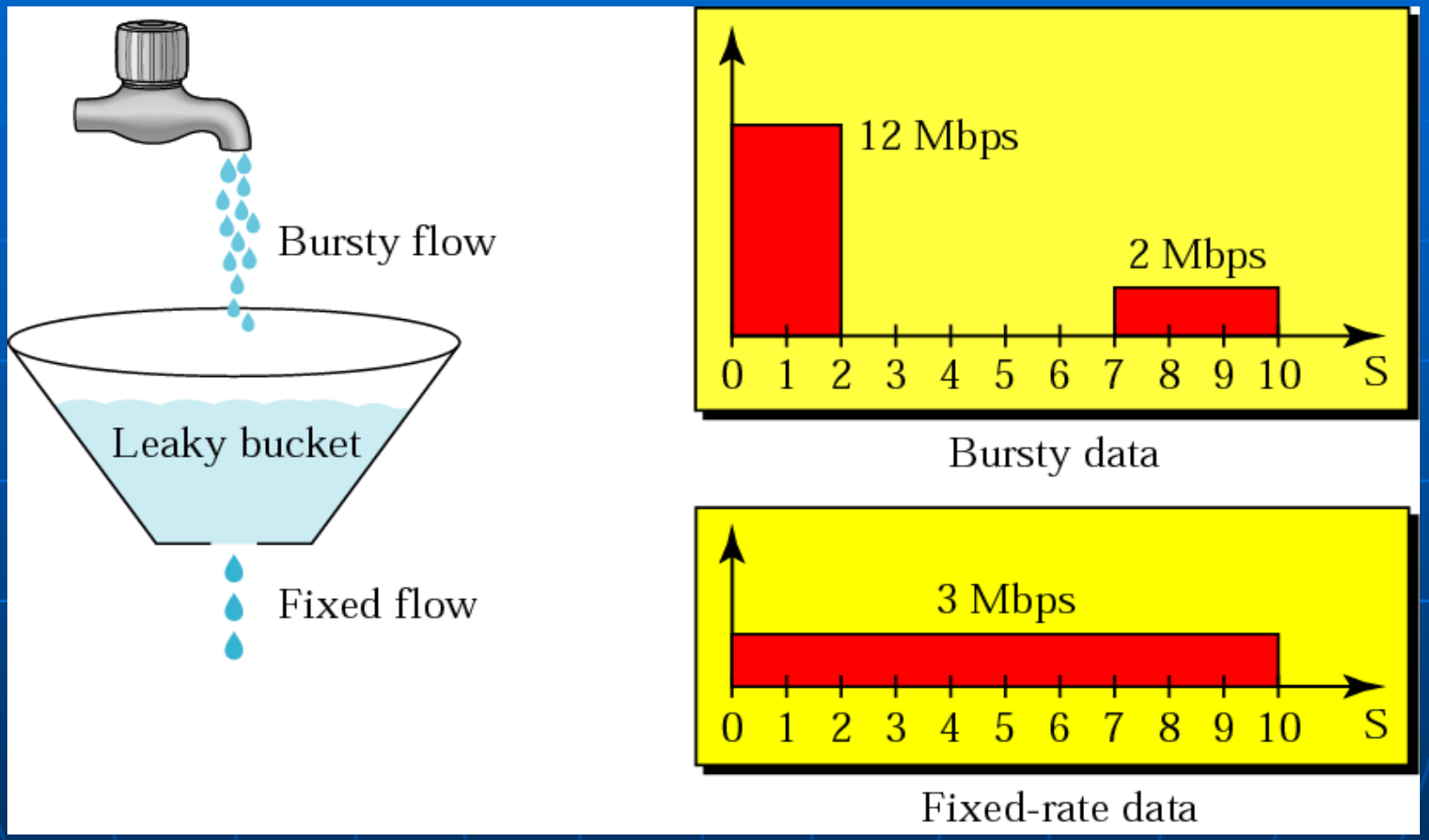
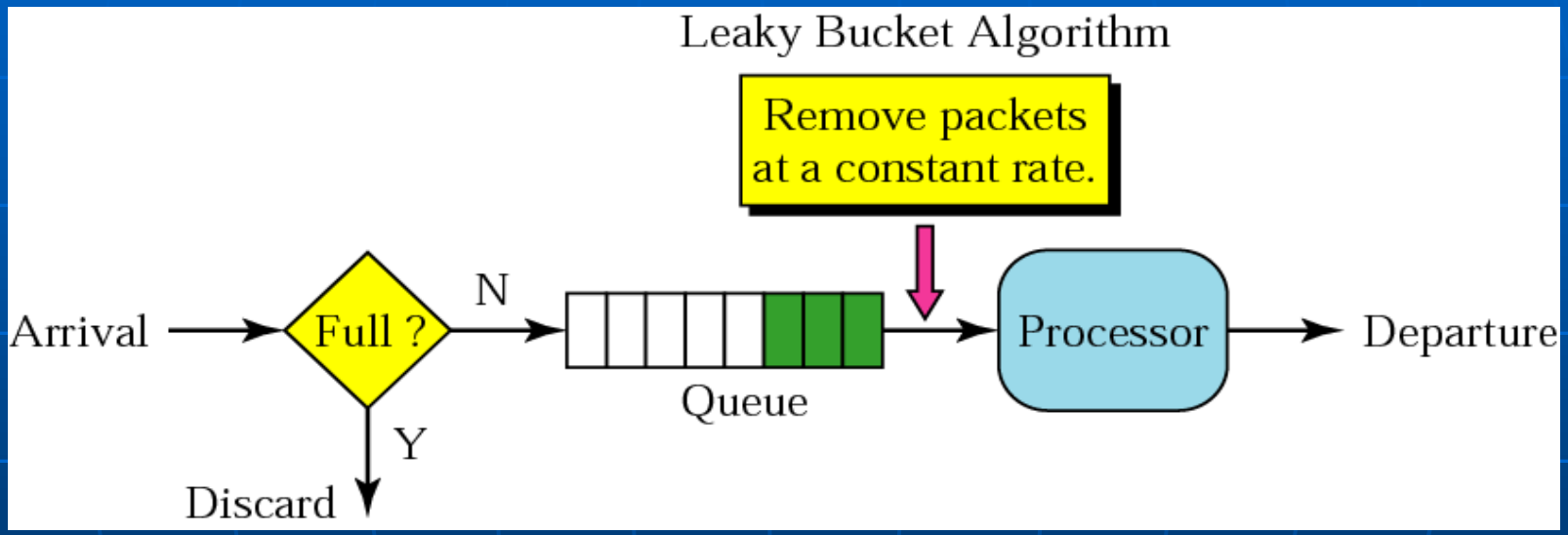
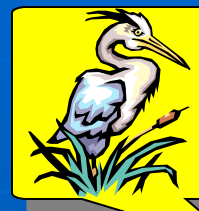


Figure 23.17 Leaky bucket implementation

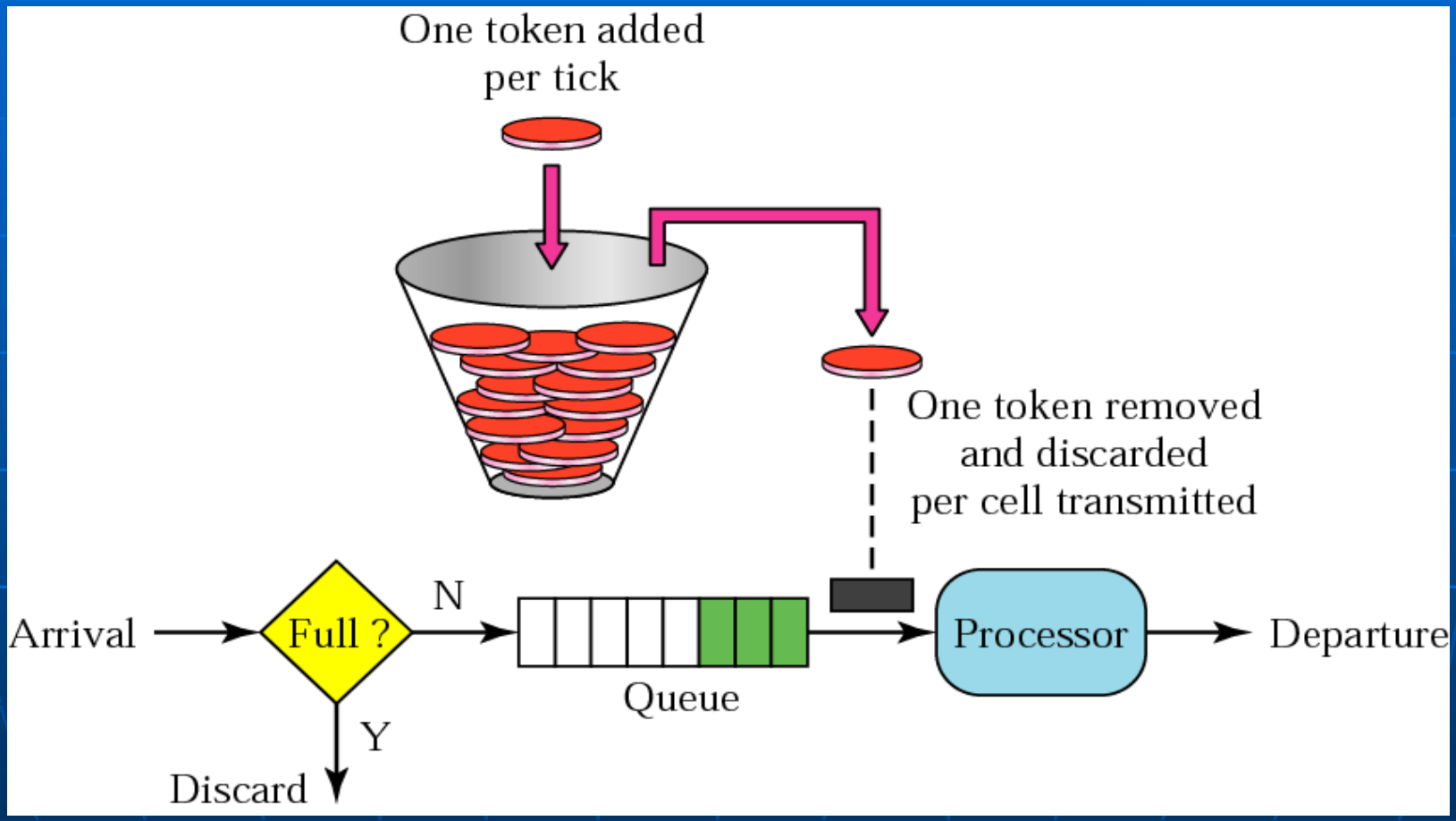




Note:

A leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging the data rate. It may drop the packets if the bucket is full.

Figure 23.18 Token bucket





The token bucket allows bursty traffic at a regulated maximum rate.

23.7 Integrated Services

Signaling

Flow Specification

Admission

Service Classes

RSVP



Integrated Services is a flow-based QoS model designed for IP.

Difference

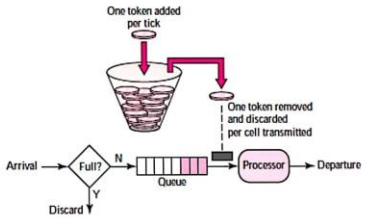
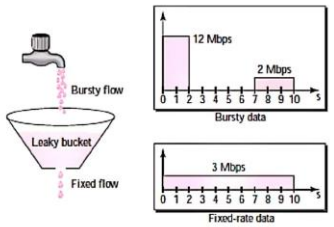
Sr no.	Token Bucket	Leaky Bucket
1	Token bucket is token dependent.	Leaky bucket is token independent.
2	Block diagram token bucket. 	Block diagram of leaky bucket . 
3	If bucket is full token are discarded but not the packet.	If bucket is full packet or data is discarded.
4	Token bucket allows for large bursts to be sent faster by speeding up the output.	Leaky bucket sends the packets at an average rate.
5	Token bucket allows saving up of tokens (permission) to send large bursts.	Leaky bucket does not allow saving a constant rate is maintained.
6	Packets can only Transmitted when there are enough token.	Packet are transmitted continuously.
7	It save token.	It is does not save token.

Figure 23.19 Path messages

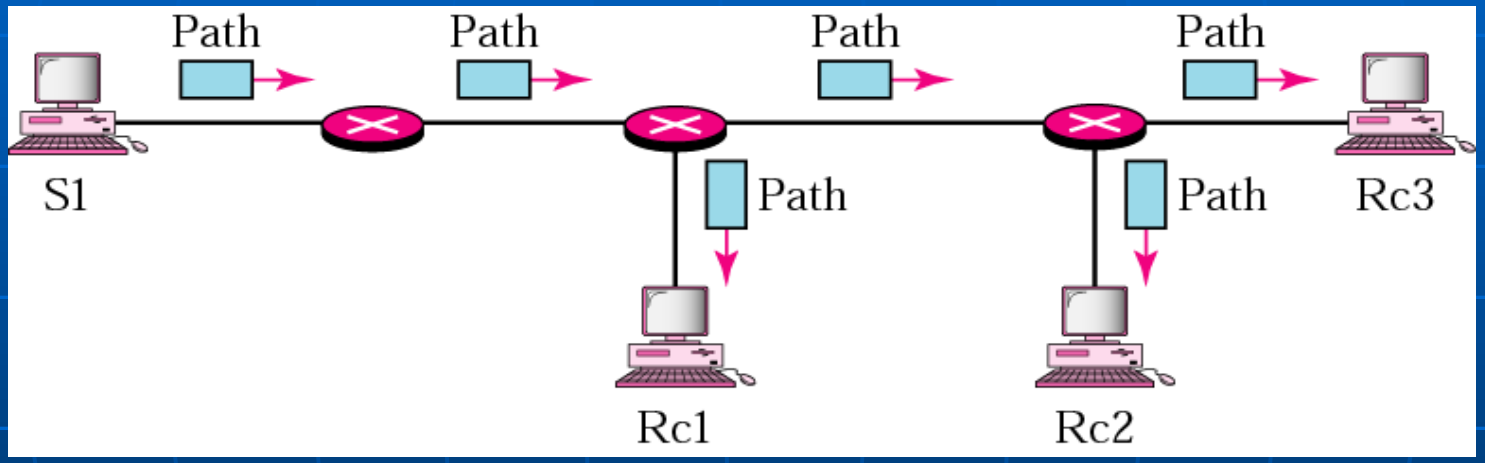


Figure 23.20 Resv messages

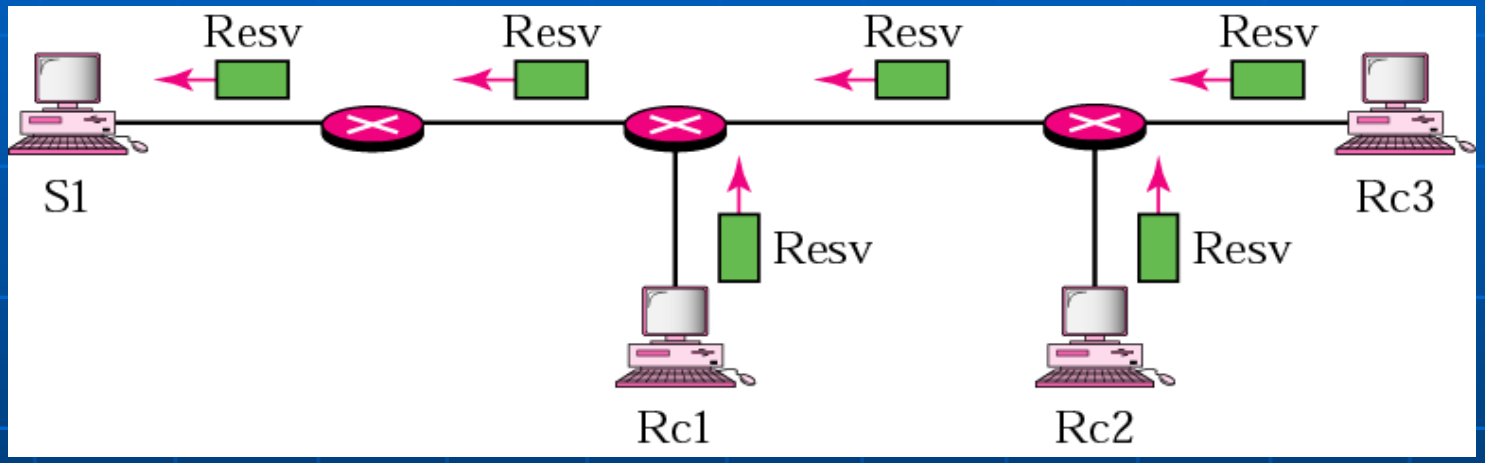


Figure 23.21 Reservation merging

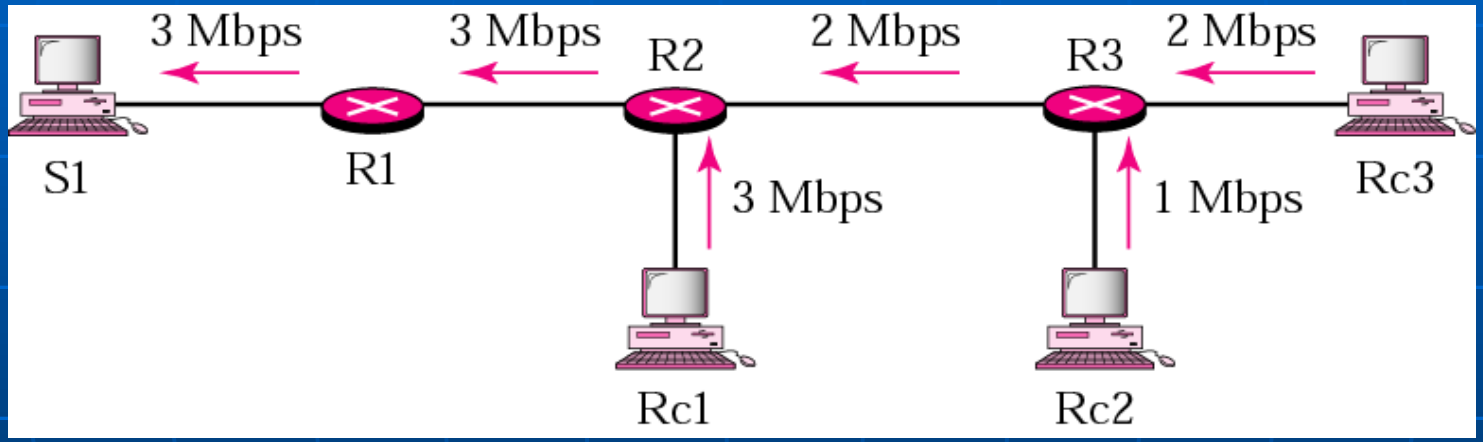
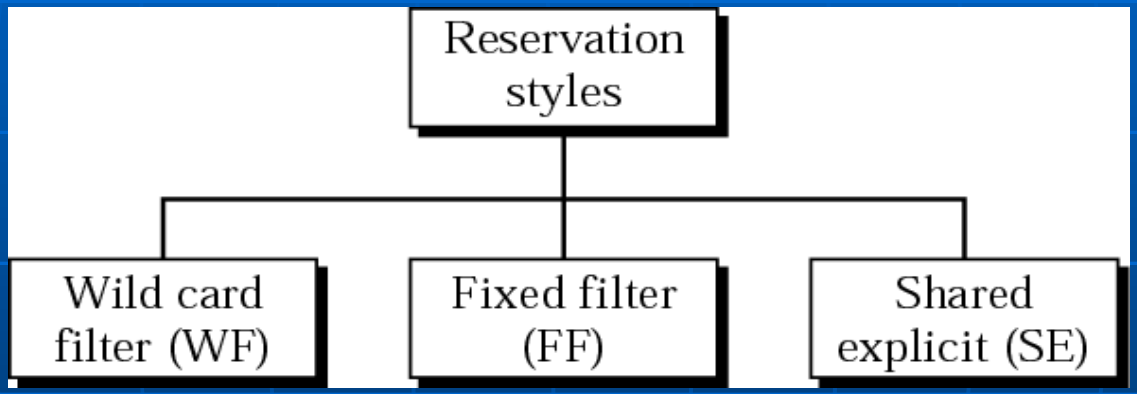


Figure 23.22 Reservation styles



23.8 Differentiated Services

An Alternative to Integrated Services



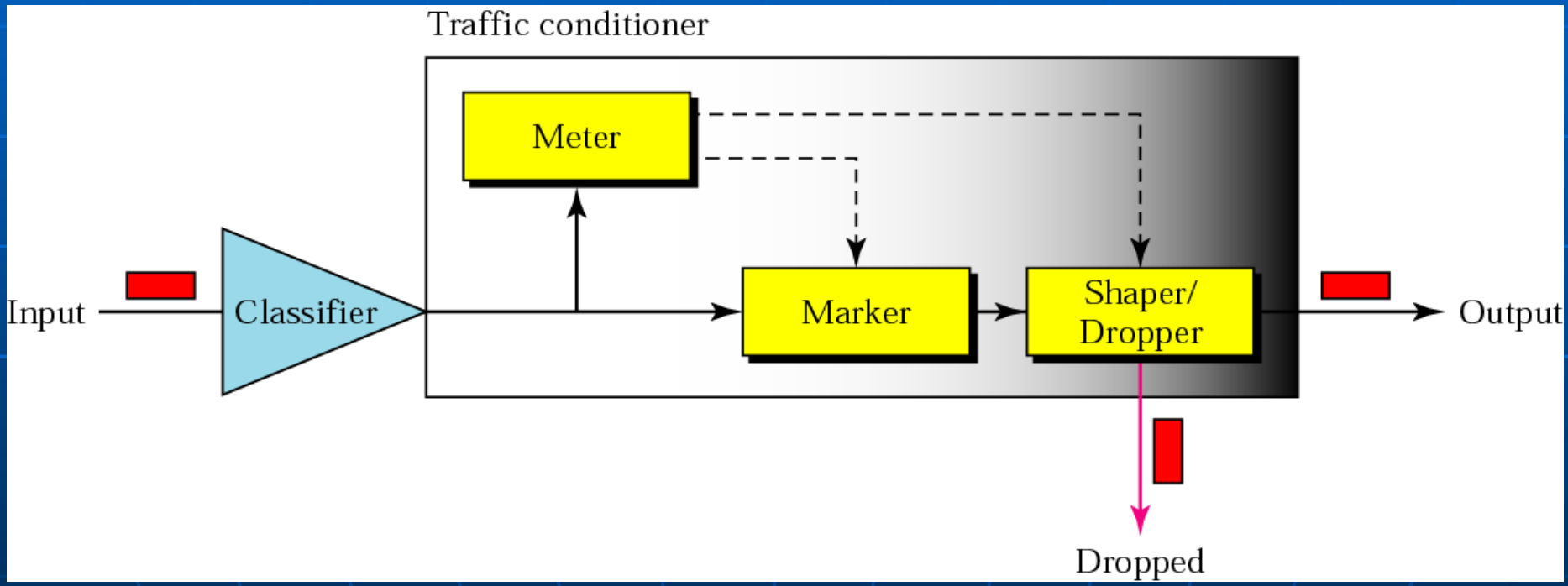
Note:

Differentiated Services is a class-based QoS model designed for IP.

Figure 23.23 DS field



Figure 23.24 Traffic conditioner



23.9 QoS in Switched Networks

QoS in Frame Relay

QoS in ATM

Figure 23.25 Relationship between traffic control attributes

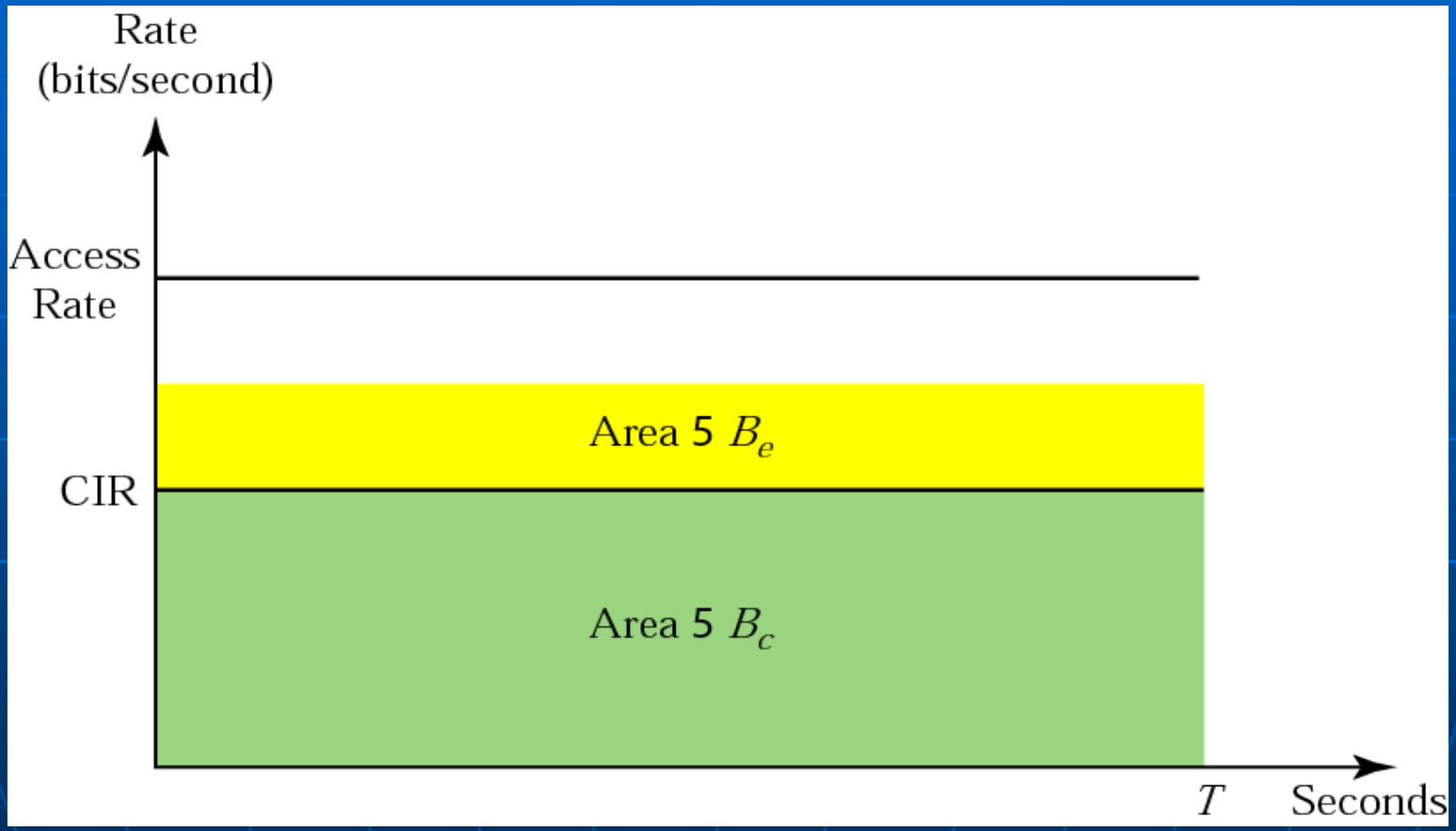


Figure 23.26 User rate in relation to Bc + Be

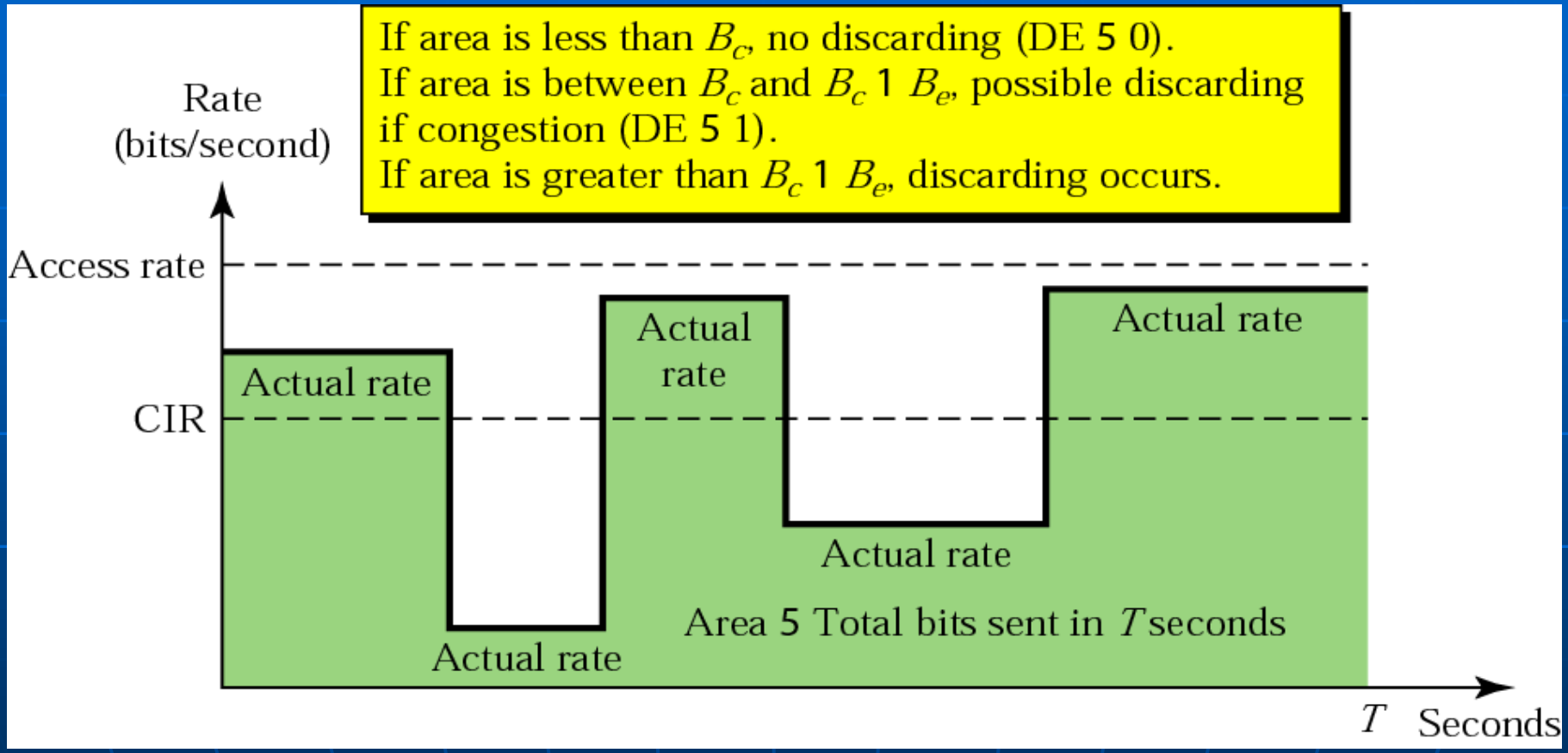


Figure 23.27 Service classes

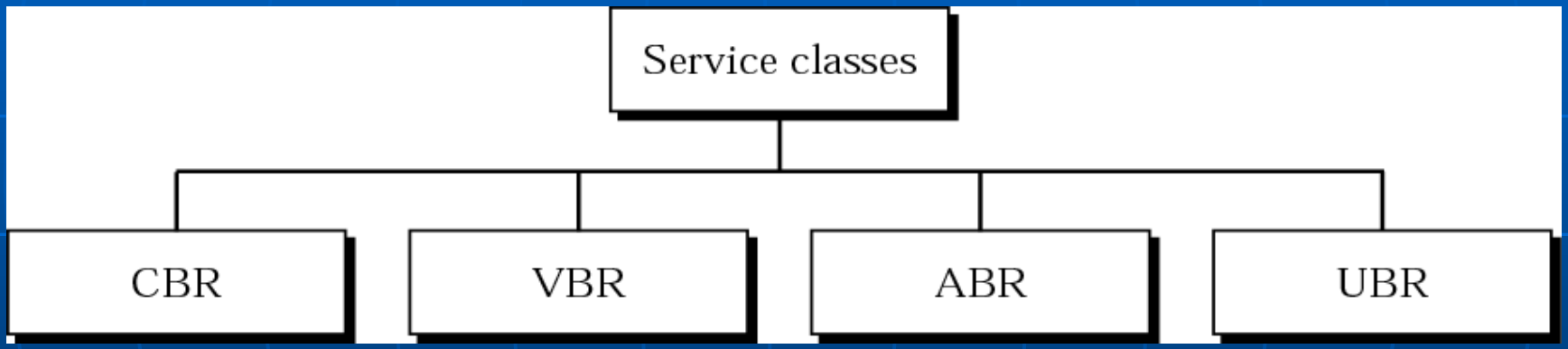


Figure 23.28 Relationship of service classes to the total capacity

